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Leveraging inter-arrival time for efficient threat filtering: A parsimonious approach

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ABSTRACT

In this study, we propose a streamlined approach to intrusion detection by leveraging the Interpacket Arrival Time (IAT) as a primary metric for identifying malicious network traffic. Our objective is to enhance the efficiency of intrusion detection systems by implementing a preliminary filtering layer that rapidly identifies easily detectable attacks, thereby reducing the computational load on more sophisticated, resource-intensive models. Using datasets such as CICIoT2023, CIC-IDS-2017, and UNSW-NB15, we conducted extensive experiments to validate the effectiveness of our approach. The study employed techniques like SMOTE to address dataset imbalances and Min-Max scaling to normalize the IAT feature, ensuring optimal performance of machine learning models. We evaluated models such as Random Forest, K-Nearest Neighbors, and Multilayer Perceptron, with a particular emphasis on their ability to generalize across various datasets. Our findings demonstrate that by focusing on a single, well-chosen feature like IAT, it is possible to achieve high detection accuracy while significantly reducing training and prediction times. This method not only improves the overall efficiency of intrusion detection systems but also suggests a practical solution for real- time applications where resource constraints are a critical concern.

1. Introduction

As technology continues to advance, the field of cybersecurity faces an ever-growing array of challenges. Intrusion detection systems (IDS) are critical tools for monitoring networks and identifying malicious behavior (Denning, 1987). Traditionally, IDSs have relied on handcrafted rules and signatures, which, while effective to some extent, lack adaptability and struggle to cope with the rapidly evolving nature of cyber threats. The advent of machine learning has significantly influenced the development of IDSs, enabling systems to learn and adapt to new attack patterns dynamically (Buczak and Guven, 2016; Laskov et al., 2005). However, the integration of machine learning into IDSs has introduced its own set of challenges. The vast amounts of network data that must be processed can make these algorithms complex and susceptible to overfitting (Chandrashekar and Sahin, 2014). Traditional machine learning methods often struggle to keep pace with the sophistication of emerging cyber threats. While these approaches have the potential to enhance detection capabilities, they also introduce issues related to computational efficiency and model interpretability, particularly when dealing with large and diverse datasets.

The increasing variety of attack types and the diversity of network environments further complicate the intrusion detection landscape (Garcia et al., 2014). These challenges highlight the need for more efficient methods that can maintain high accuracy while reducing data dimensionality and computational complexity. In this study, we propose a first-tier preliminary threat filtering system, which acts as an initial layer of defense by quickly identifying and filtering out easily detectable threats. This approach is intended to reduce the burden on more complex and resource-intensive detection models. To implement this preliminary filtering system, we focus on the interpacket arrival time (IAT) feature. The IAT has been extensively researched in the literature and has shown promise as a key indicator of network anomalies (Guyon and Elisseeff, 2003). We believed that concentrating on this feature would be a good starting point for developing an efficient intrusion detection model, and our results demonstrate that this approach is indeed effective.

This study seeks to demonstrate that concentrating on the IAT feature can improve the speed and accuracy of intrusion detection models while effectively filtering out obvious threats in the preliminary stage (Huan and Lei, 2005). Our findings underscore the value of

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focusing on well-researched features in enhancing the efficiency of IDSs, with potential applications in real-world scenarios where rapid and accurate intrusion detection is essential. Additionally, we provide a detailed analysis of the results using a 30-run cross-validation, which ensures that our evaluation is statistically robust and less prone to variability.

In the context of network security, the IAT is emphasized in our approach because it captures critical temporal dynamics in network traffic, making it an effective single feature for detecting certain classes of anomalies. In the context of network security, abnormal timing behaviors often arise as a direct result of malicious activities that disrupt normal traffic patterns. For example, denial-of-service (DoS) attacks typically flood the network with a high volume of packets over a short period, significantly reducing the time intervals between successive packets. This leads to distinct deviations in IAT, creating a detectable signature that can be used to flag potentially harmful activity. Theoretically, IAT serves as a proxy for network load, and in attack scenarios, it provides a clear indication of disruption to normal traffic flow. By isolating IAT as a key metric, we focus on the immediate temporal anomalies that are indicative of such network disruptions, which are often the earliest detectable signs of an attack.

Beyond DoS attacks, the effectiveness of IAT extends to detecting more subtle threats, such as re- connaissance activities. In these cases, attackers typically probe the network in search of vulnerabilities, generating irregular traffic patterns. These scans often occur at uneven intervals, which result in sporadic spikes in IAT values. The irregularity and unpredictability of these spikes make IAT a useful feature for identifying scanning behaviors that deviate from typical network traffic, which tends to follow smoother, more predictable timing patterns. Furthermore, when attackers attempt to exfiltrate data or perform other illicit activities, their packet timing may differ due to variations in loading speeds, packet encryption, or additional obfuscation techniques. The IAT captures these subtleties, allowing for detection even when traditional volume-based or content-based metrics fail to pick up on the intrusion.

The approach we advocate is grounded in the principle of parsimony in statistical learning, which suggests that focusing on a small, well-chosen set of features—like IAT—can yield better generalization and faster detection times in real-time systems. By reducing the feature set to IAT, we reduce the dimensionality of the data, allowing our models to operate more efficiently without losing detection power. In high-traffic environments, this focus on a single feature minimizes computational overhead and reduces model complexity, which helps mitigate issues like overfitting or unnecessary noise introduced by extraneous features. From a theoretical standpoint, this approach works well in domains where timing anomalies are early indicators of compromise, and where computational efficiency is critical. The simplicity of focusing on IAT allows for faster decision-making while still capturing the key temporal dynamics that differentiate benign from malicious traffic.

The key contributions of this study are:

- Proposing a fast first-tier preliminary threat filtering system that enhances the efficiency of intrusion detection by offloading simpler tasks from more sophisticated models.
- Providing a detailed analysis of the results, supported by a rigorous 30-run cross-validation process.
- Demonstrating the effectiveness of our approach across multiple datasets, highlighting its potential for real-world applications.
- Conducting the first extensive examination of the interpacket arrival time (IAT) feature across different datasets, offering valuable insights into its role in enhancing intrusion detection systems.

2. Threat model and practical implementation in IDS frameworks

In our proposed system, we recognize that cybersecurity best practices emphasize the necessity of multiple lines of defense. Each line acts as a barrier, designed to mitigate threats should one layer be breached. Interpacket Arrival Time (IAT) analysis serves as an essential component within this larger, multi-tiered defense strategy. While IAT is powerful in detecting specific timing-based anomalies, it is by no means intended as the sole defense mechanism. Instead, it functions alongside other countermeasures to build a holistic and resilient security framework. This layered approach ensures that, if an adversary circumvents one layer, another is in place to respond, thereby increasing the overall robustness and resilience of the network.

Additionally, similar to many cybersecurity strategies, our system incorporates a careful balance between security and performance. IAT functions as a "first line of defense," effectively filtering out the most obvious, lower-level threats—such as rapid packet bursts typical of DDoS attacks—without significant computational overhead. This initial, quick-pass approach enables the system to conserve resources, allowing more compute- intensive defenses to focus on the more sophisticated threats that demand their attention. By prioritizing the detection of anomalies with minimal processing, we prevent unnecessary load on the network and ensure that more complex security layers remain agile and effective. This methodology resembles early rejection processes used in academic peer review or triage strategies in resource management, avoiding "overkill" responses and enhancing the efficiency of the entire system.

Our system's design, focused on packet timing rather than packet content, is particularly beneficial in environments with encrypted traffic, where content inspection offers limited visibility. By capturing timing anomalies, our approach provides an additional layer of defense that functions effectively, even when packet contents are obscured. This method detects deviations in packet timing that might signal potential attacks or reconnaissance activities, even if traffic volume remains unremarkable. In high-traffic settings, this early-warning capability is crucial, as it enables rapid identification of potential threats without degrading network performance. By emphasizing temporal behavior, our system ensures continuous security vigilance, detecting and flagging suspicious timing patterns that may indicate malicious intent, thereby enhancing network defense in real-time.

2.1. Threat model and attack scenarios

The types of attacks our IAT-based detection system is designed to detect are well-represented by the datasets used in our experiments, including CICIoT2023, UNSW-NB15 and CIC-IDS-2017 datasets. These datasets cover a broad range of network threats, such as Denial-of-Service (DoS), Distributed Denial-of- Service (DDoS), brute force attacks, spoofing, reconnaissance, web-based attacks, and more. Specifically, our approach is optimized to detect anomalies in traffic patterns linked to these attack vectors by identifying timing irregularities that deviate from normal, benign network behaviors. For instance, during DDoS attacks, the high frequency of packet transmission results in shorter interpacket arrival times, making IAT a strong indicator of abnormal traffic. In our threat model, the adversary is assumed to have the capability to generate malicious traffic and possibly try to evade detection by adjusting their attack traffic's timing characteristics. This model anticipates attacks such as brute force FTP or SSH attacks, reconnaissance scans, botnet activity, and application-layer DoS attacks, all of which may exhibit timing patterns that the IAT analysis can help flag as suspicious. However, sophisticated attackers may attempt to obfuscate their IAT values to mimic normal traffic. Our IAT-based system is designed to serve as an initial line of defense to capture common and easily identifiable attacks, and further processing by more sophisticated systems will handle advanced, obfuscated threats.

2.2. Implementation in existing IDS frameworks

Integrating the IAT-based approach into existing intrusion detection

systems (IDS) such as Zeek (for- merly Bro), Suricata, and Snort provides a practical deployment strategy that enhances the overall detection pipeline by introducing an efficient preliminary filtering stage. Below, we outline how IAT can be imple- mented in real-world systems.

In the initial phase, the IAT values between incoming packets are calculated using a lightweight module added to the IDS. Zeek or Suricata, which are already well-suited for monitoring traffic, can be configured to compute and log IAT values as part of their packet analysis workflows. A threshold-based mechanism can be applied to flag packets with unusual timing intervals, which is where the IAT-based analysis would excel.

Once flagged by the IAT module, the traffic can then be passed on to more resource-intensive detection systems, such as deep packet inspection (DPI) engines in Snort or Suricata, which focus on analyzing packet contents for known attack signatures. These systems are effective in detecting sophisticated attacks but are computationally expensive. By utilizing the IAT-based system as a first-pass filter, we significantly reduce the number of packets requiring deep inspection, allowing the overall system to operate more efficiently.

The IAT detection system works by applying machine learning models trained on datasets containing real-world attacks such as DDoS, DoS, brute force, and reconnaissance scans. When an attack is detected based on timing irregularities, packets can either be classified as suspicious and flagged for further inspection by Suricata or Zeek or classified as benign, allowing them to proceed without delay. This layered approach of using IAT as an initial screening method improves both speed and precision in high-traffic networks.

The practical deployment of this IAT-based system allows it to fit seamlessly into existing IDS infrastruc- tures. The primary advantage lies in its ability to quickly filter out benign traffic and highlight suspicious packets early, effectively reducing the computational burden on more complex detection layers, such as signature-based systems in Snort or rule-based mechanisms in Zeek. This multi-tiered approach ensures that the IDS operates at peak efficiency without sacrificing detection quality.

Fig. 1 illustrates the integration of the IAT-based detection system into a real-world IDS framework. Incoming packets are first analyzed by the IAT module, which flags traffic that exhibits abnormal timing patterns. If packets are flagged as suspicious, they are passed on to more computationally intensive systems like Zeek, Suricata, or Snort for deeper analysis. Packets that are easily identifiable as benign or malicious based on IAT values can skip the advanced inspection step, reducing the load on these systems. In this way, IAT serves as a lightweight filter that optimizes overall system performance.

In conclusion, integrating the IAT-based approach into existing IDS frameworks provides a practical solution for real-world deployment. This method serves as an efficient preliminary layer that reduces the processing load on more sophisticated models, allowing intrusion detection systems to operate with greater speed and efficiency while maintaining high detection accuracy. The effectiveness of this layered approach in detecting common attacks such as DDoS, brute force, reconnaissance, and spoofing attacks makes it a valuable addition to any IDS infrastructure.

3. Literature review

3.1. General approaches and observations in intrusion detection

He et al. provide a comprehensive survey on the application of adversarial machine learning within Net- work Intrusion Detection Systems (NIDS), highlighting the vulnerabilities of deep neural networks (DNNs) to adversarial attacks. The study discusses how these attacks, often developed for computer vision tasks, are being adapted to exploit the detection pipelines and feature spaces of NIDS. The authors categorize various adversarial attack methods, such as white-box and black-box approaches, and evaluate their effectiveness in the NIDS domain. Additionally, the survey reviews current defense mechanisms against these attacks, offering insights into the challenges and trends in enhancing the resilience of DNN-based NIDS to adversarial manipulation (He et al., 2023).

Vasilomanolakis et al. conduct a thorough survey on Collaborative Intrusion Detection Systems (CIDS), which have emerged as a scalable alternative to conventional IDSs in the face of increasingly sophisticated cyber threats. Their work outlines the fundamental requirements for CIDS, and introduces a comprehensive taxonomy of its various components, including data collection, analysis, and alert correlation mechanisms. The survey also discusses the potential attacks that can evade CIDS and those targeting the availability of the systems themselves. This detailed examination of the state-of-the-art in CIDS provides valuable insights into the design and implementation of collaborative approaches for enhanced network security (Vasilomanolakis et al., 2015).

Zarpelão et al. provide an extensive survey of Intrusion Detection Systems (IDS) tailored for the Internet of Things (IoT), addressing the unique challenges posed by IoT environments such as constrained-resource devices and specialized protocol stacks. The paper classifies IDS approaches based on detection methods, placement strategies, security threats, and validation strategies, offering a detailed analysis of existing research efforts. This survey identifies key trends and highlights open issues, paving the way for future research directions in the application of IDS to IoT (Zarpelão et al., 2017).

Drewek-Ossowicka et al. present a comprehensive survey on the application of neural networks (NN) in Intrusion Detection Systems (IDS), highlighting the significant advancements in AI and their impact on network security. The paper provides an overview of recent literature, including both surveys and novel method proposals, demonstrating the increasing reliance on neural networks to enhance IDS performance. Additionally, the authors offer brief tutorials on neural network architectures, types of IDS, and relevant training datasets, making this survey a valuable resource for understanding the current state and future

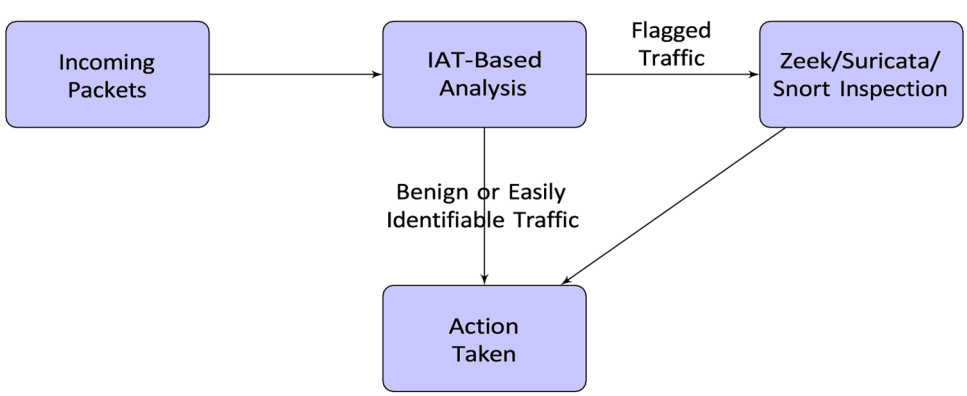

Fig. 1. IAT-Based Detection Workflow with Advanced IDS Integration.

potential of NNs in intrusion detection (Drewek-Ossowicka et al., 2021).

Khraisat et al. provide an extensive survey on Intrusion Detection Systems (IDS), categorizing them into Signature-based (SIDS) and Anomaly-based (AIDS) techniques. The paper reviews notable recent developments in the field, discusses commonly used datasets for IDS evaluation, and explores the evasion techniques employed by attackers to bypass detection. This survey offers valuable insights into the challenges facing IDS research, particularly in adapting to increasingly sophisticated cyber-attacks, and highlights potential directions for future work aimed at enhancing the security of computer systems (Khraisat et al., 2019).

Sahani et al. present an in-depth survey of machine learning-based Intrusion Detection Systems (IDS) specifically tailored for smart grid environments. The survey covers key aspects including the application of ML-based IDS in the transmission and distribution components of smart grids, dataset generation, and the evaluation metrics and testbeds used. The authors also discuss the unique security vulnerabilities of smart grids and provide insights into future research directions to enhance IDS effectiveness in these critical infrastructures (Sahani et al., 2023).

Momand et al. provide a systematic and comprehensive survey of recent advances in Intrusion Detection Systems (IDS) with a focus on machine learning, particularly deep learning techniques. The paper discusses various IDS methodologies, the evolution of datasets used for training these systems, and offers a taxonomy of attack types. Additionally, the authors highlight future research directions and challenges in detecting and mitigating malicious activities across different network environments (Asadullah et al., 2023).

3.2. Studies utilizing time-based metrics in IDS research

Su et al. propose the BAT model, a deep learning-based approach for network intrusion detection, ad- dressing the limitations of traditional machine learning methods like KNN and SVM, which often suffer from low accuracy and a heavy reliance on manual feature engineering. The BAT model integrates Bidirectional Long Short-Term Memory (BLSTM) with an attention mechanism, enhancing the extraction of key features from network traffic data for more effective classification. Tested on the NSL-KDD dataset, the BAT model demonstrates superior performance compared to other methods, making it a significant advancement in the field of intrusion detection (Su et al., 2020).

Liu et al. explore the application of machine learning techniques for intrusion detection in Contiki-NG- based IoT networks using the NSL-KDD dataset. The study evaluates eleven machine learning algorithms, finding that tree-based methods and ensemble techniques, particularly XGBoost, perform exceptionally well in detecting network intrusions. The paper also notes the strong performance of the unsupervised Expectation-Maximization (EM) algorithm, which surpasses the accuracy of the Näive Bayes classifier by a significant margin, highlighting its potential for intrusion detection in IoT environments (Liu et al., 2020).

Sarker et al. introduce the IntruDTree model, a machine learning-based intrusion detection system de- signed to enhance cybersecurity by focusing on the importance of feature selection. The model prioritizes security features based on their relevance, constructing a tree-based intrusion detection framework that not only improves prediction accuracy but also reduces computational complexity by minimizing feature dimen- sions. The effectiveness of the IntruDTree model is validated through extensive experiments on cybersecurity datasets, where it outperforms traditional machine learning methods like Näive Bayes, logistic regression, SVM, and k-nearest neighbor in key performance metrics such as precision, recall, F-score, accuracy, and ROC values (Sarker et al., 2020).

Alqahtani et al. investigate the application of various machine learning classification techniques for cyber intrusion detection, focusing on algorithms such as Bayesian Network, Naive Bayes, Decision Tree, Random Decision Forest, Random Tree, Decision Table, and Artificial Neural Networks. The study assesses these methods using cybersecurity datasets, analyzing their effectiveness across multiple performance metrics, including precision, recall, F1-score, and accuracy. The results highlight the potential of machine learning techniques in enhancing intrusion detection systems by providing intelligent, data-driven security solutions (Alqahtani et al., 2020).

Abrar et al. explore the application of various machine learning classifiers, including Support Vector Ma- chine (SVM), K-nearest neighbor (KNN), logistic regression (LR), Näive Bayes (NB), Multi-layer Perceptron (MLP), random forest (RF), Extra-tree classifier (ETC), and decision tree (DT), for the development of an intrusion detection system (IDS) using the NSL-KDD dataset. Their research highlights the importance of feature selection and preprocessing in enhancing IDS performance, with empirical results showing that RF, ETC, and DT achieved over 99 % accuracy across all attack classes. The study demonstrates that the proposed model not only effectively predicts intrusions but also reduces computational complexity by eliminating irrelevant features (Abrar et al., 2020).

Asif et al. propose the MapReduce-Based Intelligent Model for Intrusion Detection (MR-IMID), which integrates machine learning techniques with the MapReduce framework to efficiently handle the large vol- umes of data generated by modern networks, particularly in the context of the Internet of Things (IoT). The MR-IMID model is designed to automate intrusion detection by performing multiple data classification tasks across distributed computing resources. The study demonstrates that MR-IMID achieves high detec- tion accuracy, with 97.7 % during training and 95.7 % during validation, outperforming previously published approaches and offering a scalable solution for real-time intrusion detection in large-scale networks (Asif et al., 2022).

3.3. Studies focused exclusively on time-based metrics in IDS research

Moore et al. present a novel approach to in-vehicle intrusion detection by modeling inter-signal arrival times on the Controller Area Network (CAN) bus to accurately detect signal injection attacks. The study highlights the growing vulnerability of modern vehicles due to the increasing connectivity of electronic control units (ECUs) and external interfaces. The proposed data-driven anomaly detection algorithm is effective in identifying anomalous traffic patterns based on unusual refresh rates of CAN bus signals. The algorithm achieves a high true positive rate of 0.9998 and a low false discovery rate of 0.00298, demonstrating its efficacy in real-world scenarios (Moore et al., 2017). The study's limitations include its reliance on CAN bus traffic data gathered from a single, undisclosed vehicle's make, model, and year, focusing on specific Process IDs (PIDs), which may not generalize across different vehicles or attack types without further adaptation and testing.

Song et al. propose a light-weight intrusion detection system (IDS) for in-vehicle networks, specifically targeting the Controller Area Network (CAN) bus, which is vulnerable to both internal and external attacks due to its lack of inherent security features. The authors' approach analyzes the time intervals of CAN messages to detect anomalous behavior indicative of signal injection attacks. The IDS was tested on real CAN traffic data, successfully detecting all injection attacks without generating false positives, demonstrating the effectiveness of time interval analysis as a feature for in-vehicle network security (Song et al., 2016). While they achieved 100 % accuracy in detecting message injection attacks with no false positives, a limitation is that their dataset was specific to a single vehicle ("K-car"), which may not generalize across different vehicle models or broader attack scenarios.

Blevins et al. present a benchmark study on time-based intrusion detection systems (IDSs) for Controller Area Networks (CANs) in vehicles, using the newly published ROAD dataset. The study compares four time- based IDSs, highlighting the superior performance of "distribution-agnostic" methods over "distribution- based" methods in detecting message injection attacks. The authors emphasize the importance of using high-fidelity datasets with real, stealthy attacks to properly

evaluate IDS performance. Additionally, they introduce a lightweight hardware solution for deploying the best-performing IDS in a wide range of vehicles, providing a practical approach to enhancing vehicle cybersecurity (Blevins et al., 2021). While they achieved high accuracy with an F1 score of 99.0 % and an AUC-PR of 87.63 % in detecting CAN bus attacks using the ROAD dataset, a limitation is that the dataset only includes data from a single vehicle, which may not generalize across different vehicle models or broader attack scenarios.

Arshadi and Jahangir investigate the distribution of inter-arrival times of TCP flows in network traffic, finding that in "normal traffic" conditions, these inter-arrival times conform to the Weibull distribution. The study highlights that deviations from this distribution can indicate anomalies, providing a basis for a straightforward anomaly detection method. The authors use the median-rank method to estimate the parameters of the Weibull distribution and employ a chi-square test to determine the normality of the traffic, offering an effective approach to identify irregularities in network behavior (Arshadi and Jahangir, 2011). While the method effectively detects anomalies by modeling TCP flow inter-arrival times with a Weibull distribution, a limitation is that it primarily focuses on detecting deviations in traffic patterns related to SYN packets, which may not capture more sophisticated or subtle anomalies that do not significantly alter inter-arrival times.

3.4. Building upon and diverging from existing research

The existing body of research on intrusion detection systems (IDS) has explored a wide array of metrics and methodologies to enhance the detection of malicious activities. Studies such as those by Moore et al. (Moore et al., 2017) and Song et al. (Song et al., 2016) have demonstrated the effectiveness of time-based metrics, particularly inter-signal and inter-message arrival times, in detecting specific types of attacks, such as signal injection attacks within in- vehicle networks. These approaches emphasize the importance of temporal patterns in identifying abnormal behaviors that are indicative of an intrusion. However, these studies are often confined to highly specialized environments, such as CAN bus systems in vehicles, where the attack vectors and network conditions are vastly different from those in broader network infrastructures like the Internet of Things (IoT) or general enterprise networks.

Our work builds upon these foundational studies by extending the application of interpacket arrival time (IAT) as a critical metric for intrusion detection beyond these specialized environments. Unlike the in-vehicle network studies that focus on specific message types and controlled environments, our approach is designed for more diverse and dynamic network scenarios. By applying IAT in a broader context, we explore its potential as a universal metric for early threat filtering across various datasets, including CICIoT2023, CIC-IDS-2017, and UNSW-NB15. This broad application is a key divergence from existing research, where the focus has typically been on either multi-metric approaches or environments with highly predictable traffic patterns.

Moreover, while previous studies have often integrated IAT with other features to enhance detection accuracy, our research diverges by testing the hypothesis that IAT alone can serve as an effective metric for preliminary filtering. This "single-feature" approach is grounded in the observation that IAT can capture significant anomalies indicative of various types of attacks, particularly those that involve rapid or irregular packet transmission. Our findings suggest that using IAT as a standalone metric allows for the development of a fast and lightweight detection mechanism, which can offload the burden from more complex models and improve overall system efficiency. This is particularly valuable in real-time applications where computational resources and response times are critical. Thus, our work not only builds upon the existing understanding of time-based metrics in IDS but also introduces a novel application of IAT, paving the way for more streamlined and efficient intrusion detection strategies.

4. Methodology

4.1. Utilized datasets

In this study, we employed three distinct datasets to evaluate the effectiveness and generalizability of our intrusion detection methodology: CICIoT2023, CIC-IDS-2017, and UNSW-NB15. Each of these datasets offers unique characteristics and challenges, allowing us to comprehensively assess the performance of our proposed models across different network environments and types of cyberattacks.

The CICIoT2023 dataset is particularly valuable for its focus on IoT environments, capturing a wide range of attacks in a realistic smart home setting. This dataset provides a robust testbed for evaluating intrusion detection methods in scenarios that closely resemble real-world IoT deployments. The CIC-IDS- 2017 dataset is a comprehensive collection of various types of network traffic, including both normal and malicious activities. It is well-known for its detailed representation of modern network traffic, encompassing a wide array of attack types, which makes it an excellent benchmark for testing intrusion detection systems. The UNSW-NB15 dataset, another widely used benchmark, includes both synthetic and real network traffic, providing a diverse and challenging dataset for evaluating intrusion detection algorithms. It covers a variety of attack vectors and normal activities, ensuring that the models are tested against a broad spectrum of potential network intrusions. By using these three datasets, we ensure that our models are tested against a diverse set of network conditions and attack scenarios, enabling us to draw robust conclusions about their effectiveness and applicability in different contexts.

4.1.1. CICIoT2023

The CICIoT2023 dataset is a comprehensive and real-time collection designed by Neto et al. (Neto et al., 2023) to simulate a realistic Internet of Things (IoT) environment. This dataset encompasses a network topology consisting of 105 devices, capturing a wide range of network traffic scenarios, including both benign traffic and various types of cyberattacks. It is particularly valuable for evaluating intrusion detection methods in scenarios that closely resemble real-world IoT deployments. The dataset includes 47 features that describe different aspects of network traffic. These features provide detailed insights into packet behavior, offering a nuanced understanding of the network's operation. Key features include fundamental attributes such as flow duration, IAT, header length, protocol types, and various TCP/IP flag values, which are crucial for understanding the basic structure and flow of network packets.

In addition to these fundamental attributes, the dataset captures statistical measurements such as the minimum, maximum, average, and standard deviation of packet lengths. These metrics are vital for analyzing the variability and consistency in packet sizes, which can indicate normal versus abnormal network behavior. The dataset also contains information on the rate of packet transmission within a flow, distinguishing between outbound and inbound packets. Specific indicators for various protocols, such as HTTP, HTTPS, DNS, Telnet, and others, are included, helping to identify the types of communication occurring within the network. In our study, we utilized the feature "IAT" (Interpacket Arrival Time). The "IAT" measures the time difference between consecutive packets within a flow, which makes it especially useful for identifying patterns indicative of network attacks. This feature plays a crucial role in providing a granular analysis of network traffic, helping to differentiate between normal and malicious behaviors. By leveraging the "IAT," we can gain valuable insights into the temporal dynamics of the traffic, which often serve as key indicators of various cyber threats.

Moreover, derived metrics like magnitude, radius, covariance, and variance related to the lengths of incoming and outgoing packets are part of the dataset. These derived features are particularly useful in detecting patterns and correlations within the traffic data, which can be

critical for identifying more subtle forms of cyberattacks. The CICIoT2023 dataset encompasses 34 distinct classes, of which 33 represent various attack types, while one class corresponds to benign traffic. It is important to note that this is an imbalanced dataset, with some attack types being far more prevalent than others. This imbalance reflects real-world scenarios where certain types of attacks are more common, posing additional challenges for developing machine learning models that need to perform well across all classes. The CICIoT2023 dataset offers a rich and varied set of features that can be used to build and evaluate intrusion detection systems, particularly in the context of IoT networks. Its comprehensive nature, realistic attack scenarios, and the inherent class imbalance make it an essential resource for advancing research in cybersecurity.

4.1.2. CIC-IDS-2017

The CIC-IDS-2017 dataset is a comprehensive intrusion detection dataset created by the Canadian Institute for Cybersecurity at the University of New Brunswick (Sharafaldin et al., 2018). This dataset was designed to address the limitations of previous intrusion detection datasets by providing a rich and diverse collection of network traffic data that closely resembles real-world scenarios. The CIC-IDS-2017 dataset includes a mixture of benign traffic and the most up-to-date common attack types. The attacks captured in this dataset cover a wide range of threats, including Brute Force FTP, Brute Force SSH, Denial of Service (DoS), Distributed Denial of Service (DDoS), Heartbleed, Web Attacks, Infiltration, and Botnet activities. These attacks were executed within a controlled environment to ensure realistic and consistent traffic patterns.

One of the key strengths of the CIC-IDS-2017 dataset is its emphasis on generating realistic background traffic. The dataset includes traffic generated using a B-Profile system, which models the behavior of 25 different users interacting with network services such as HTTP, HTTPS, FTP, SSH, and email protocols. This approach ensures that the benign traffic in the dataset closely mirrors actual human activity on a network. The dataset consists of labeled network flows, which include detailed information on each packet's characteristics, such as the source and destination IPs, ports, protocols, and the specific type of attack (if any). For this study, we combined all the days' worth of data into a single DataFrame, which allowed us to analyze the dataset as a whole without segmenting by individual days.

Among the numerous features provided in the dataset, we specifically utilized the "Flow IAT Mean" as our primary IAT (Interpacket Arrival Time) feature. The "Flow IAT Mean" represents the average time between packets within a single flow, and it is particularly useful for detecting patterns indicative of network attacks. This feature allows for a granular analysis of network traffic, helping to distinguish between normal and malicious behavior.

To address the issue of class imbalance, where benign traffic significantly outnumbers malicious traffic, we employed a balancing technique. Instead of using Synthetic Minority Over-sampling Technique (SMOTE), which was computationally expensive and time-consuming this very large dataset, we opted for a simpler approach. We retained all instances of the minority class (malicious traffic) and randomly selected an equivalent number of instances from the majority class (benign traffic). This method allowed us to create a balanced dataset, ensuring that our models were not biased towards the majority class while also significantly reducing the training time. This approach proved to be efficient for our analysis, facilitating the evaluation of various machine learning models on the dataset.

The CIC-IDS-2017 dataset provides a robust foundation for evaluating and developing intrusion detection systems. Its inclusion of diverse attack types, realistic traffic patterns, and detailed feature sets make it an invaluable resource for cybersecurity research.

4.1.3. UNSW-NB15

The UNSW-NB15 dataset was created by the Cyber Range Lab of UNSW Canberra using the IXIA PerfectStorm tool (Moustafa and Slay, 2015; Moustafa and Slay, 2016; N. Moustafa et al., 2017; N. Moustafa et al., 2017). This dataset is a hybrid of real modern normal activities and synthetic contemporary attack behaviors, captured to simulate a realistic network environment. The dataset contains 100 GB of raw traffic data, including a total of 2540,044 records, distributed across four CSV files. These records encompass nine different types of attacks: Fuzzers, Analysis, Backdoors, Denial of Service (DoS), Exploits, Generic, Reconnaissance, Shellcode, and Worms. To capture and process this data, tools like Argus and Bro-IDS were employed, and twelve different algorithms were developed to generate 49 features with corresponding class labels. These features are comprehensively described in the UNSW-NB15 features.csv file, which provides detailed insights into each feature used for network traffic analysis.

For our study, we specifically utilized the "sinpkt" feature, which represents the average time between the start of a flow and the first packet received. The "sinpkt" feature is particularly useful in detecting abnormal patterns of network behavior, such as those associated with reconnaissance and exploitation activities. By analyzing this feature, we can gain a deeper understanding of how attackers initiate connections and establish flows, which is critical for early-stage intrusion detection. The UNSW-NB15 dataset is widely recognized for its rich set of features and comprehensive coverage of attack types, making it a valuable resource for developing and testing network intrusion detection systems. Its blend of real and synthetic data ensures that it closely mirrors contemporary network environments, offering a robust foundation for research in cybersecurity.

4.2. Data preprocessing

In the initial stage of our preprocessing strategy, we relabeled all instances of attacks across all datasets as 1 and all benign traffic as 0. The rationale behind this binary labeling approach is rooted in our primary objective: to perform a preliminary examination as the first line of defense. By simplifying the classification problem to a binary task, we can quickly identify whether an incoming packet is potentially malicious. If any attacks are detected, future work can focus on routing these suspicious packets to more sophisticated models for detailed analysis and classification.

Given that all the datasets exhibited significant class imbalances, we employed the Synthetic Minority Over-sampling Technique (SMOTE) to address this issue (excluding the CIC-IDS-2017 dataset, where we used a different balancing method). SMOTE is a widely recognized method for generating synthetic samples in the minority class, thereby balancing the class distribution and improving the performance of machine learning models trained on imbalanced datasets (Chawla et al., 2002). By using SMOTE, we ensured that our models would not be biased towards the majority class, allowing for more accurate detection of attacks.

After addressing the class imbalance, we performed feature selection by retaining only the Interpacket Arrival Time (IAT) feature. This decision was informed by our extensive literature review, which highlighted the efficacy of IAT as a strong indicator of network anomalies and attacks. The IAT feature is particularly useful because it captures the timing between consecutive packets, revealing irregular traffic patterns often associated with malicious activities. By focusing solely on IAT, we aimed to reduce the complexity of the model while maintaining a high level of detection accuracy.

To ensure that the IAT feature was appropriately scaled for input into the machine learning models, we applied a Min-Max scaling technique. The Min-Max Scaler was chosen because it transforms the values of the IAT feature by scaling them within a fixed range, between 0 and 1. This normalization process is crucial as it ensures that the IAT values are on a consistent scale, which is essential for the model to interpret the data accurately. By normalizing the IAT values, we enhanced the ability of the machine learning algorithms to learn effectively from the data,

leading to more reliable and robust predictions.

Fig. 2 illustrates the sequential steps involved in the data preprocessing workflow for preparing the dataset for machine learning models. The process begins with the raw data, which is first subjected to binary labeling, where all instances of attacks are labeled as '1' and benign traffic as '0'. This binary classification simplifies the problem for initial detection purposes. Next, SMOTE (Synthetic Minority Oversampling Technique) is applied to handle class imbalance, ensuring that the machine learning models are trained on a more balanced dataset. Following this, feature selection is performed, where only the Interpacket Arrival Time (IAT) feature is retained, based on its effectiveness in detecting network anomalies. The selected feature is then normalized using Min-Max Scaling, which adjusts the values to a consistent range, making them suitable for input into machine learning models. The final preprocessed data is thus ready to be sent to the machine learning models for further analysis and detection of potential threats.

4.3. Experimental setup

We conducted all experiments on Google Colab, utilizing its default computing environment, which includes an Intel Xeon CPU with 2 virtual CPUs (vCPUs) and 13GB of RAM. This setup provided sufficient computational resources to efficiently run our models without requiring specialized hardware like GPUs. For model evaluation, we employed a 70–30 training-testing split, which was randomly generated for each of the 30 iterations, ensuring that the split differed every time. This randomization allowed us to capture a broader range of variability and provided a more robust evaluation of model performance. By averaging the performance metrics across these 30 runs, we minimized the influence of any single train-test split, resulting in statistically significant and reliable estimates of model performance across all datasets. This comprehensive approach ensured that our models' ability to generalize to unseen data was thoroughly assessed.

In terms of data preprocessing, as previously detailed, we simplified the datasets by converting all attack labels to 1 and all benign labels to 0. We also employed SMOTE to handle class imbalances and retained only the Interpacket Arrival Time (IAT) feature, which was normalized using Min-Max scaling. These steps were crucial to preparing the data for effective training and testing of our machine learning models. The models were evaluated using their default settings, providing a baseline for performance that can be further optimized in future work.

4.3.1. Evaluation metrics

To assess the performance of the models, we employed several standard evaluation metrics. These include:

- **Accuracy**: The proportion of correctly classified instances out of the total instances. It provides a general sense of the model's overall effectiveness.
- **F1 Score**: The harmonic mean of precision and recall, which balances the trade-off between them. It is particularly useful when dealing with imbalanced datasets, as it gives a single metric that reflects both the model's precision and recall.
- **Recall (Sensitivity)**: The proportion of true positive instances out of the total actual positives. This metric is crucial in scenarios where identifying all positive cases (e.g., all attacks) is more important than minimizing false positives.
- **Precision**: The proportion of true positive predictions out of all positive predictions made by the model. High precision indicates a low false positive rate.
- **False Positive Rate (FPR)**: The proportion of negative instances (benign) that are incorrectly classified as positive (malicious) out of all actual negative instances. A lower FPR indicates that the model is better at distinguishing benign traffic from attacks, reducing the likelihood of false alarms. This metric is crucial in environments where minimizing false positives is essential to prevent unnecessary alerts and disruptions.
- **ROC AUC (Receiver Operating Characteristic Area Under the Curve)**: A metric that measures the ability of the model to distinguish between classes. A higher AUC value indicates a better model performance in terms of its ability to discriminate between positive and negative classes.
- **Training Time**: The average time taken to train the model across all 30 runs. This metric is critical in evaluating the model's efficiency, especially when considering real-time applications.
- **Prediction Time**: The average time taken to make a prediction on the test data, measured across all 30 runs. This is a key metric for assessing the model's suitability for deployment in environments where rapid decision-making is essential.

By averaging the metrics across 30 runs, we ensured a robust evaluation of model performance, reducing the impact of variability that could arise from a single train-test split. This comprehensive approach allowed us to gain insights not only into the average performance of the models but also into the consistency and reliability of their predictions.

4.3.2. Machine learning models

In our study, we employed a diverse range of machine learning models to explore their effectiveness in the context of preliminary threat filtering. The models selected for this analysis include K-Nearest Neighbors (KNN), Weighted K-Nearest Neighbors (W-KNN), Random Forest Classifier (RFC), Multilayer Perceptron (MLP), Decision Tree Classifier (DTC), Deep Learning (DL), and Convolutional Neural Networks (CNN). Each of these models offers unique advantages and trade-offs in terms of complexity, computational efficiency, and accuracy. Our

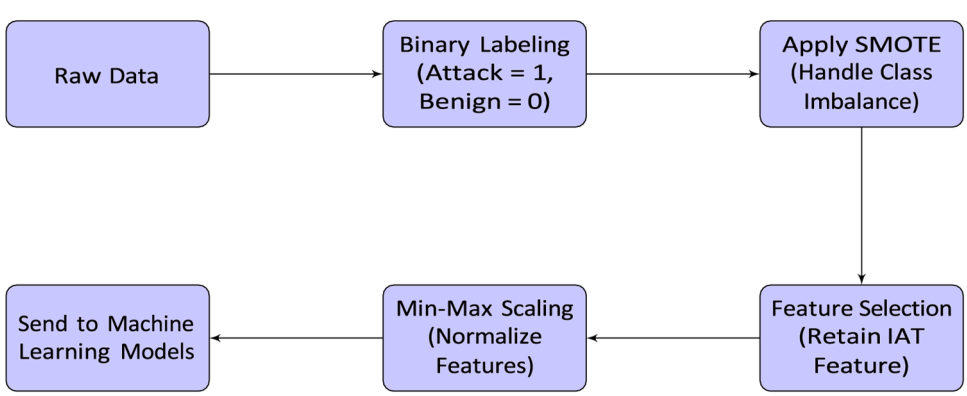

Fig. 2. Data Preprocessing Workflow.

goal was to evaluate these models to determine their suitability for rapid, first-tier filtering of network traffic based on interpacket arrival time (IAT). Notably, we excluded Support Vector Machines (SVM) from our analysis. During our experiments, we found that SVM was unable to train within a reasonable timeframe on our large-scale datasets, with the process taking several hours without completion. This excessive computational demand is contrary to the purpose of our "fast preliminary threat filtering" approach. As a result, SVM was deemed impractical for this specific task, reinforcing our focus on more efficient and scalable algorithms. Below are all of the models that we used:

K-Nearest Neighbors (KNN) (Cover and Hart, 1967):

- **Explanation**: KNN is a non-parametric method that classifies samples based on the majority vote of their nearest neighbors. It is straightforward and relies on the assumption that similar data points (neighbors) exist in close proximity to one another.
- **Pros**: KNN is simple to implement and requires no assumptions about the underlying data distribution. It is particularly useful for quick, preliminary filtering where the goal is to identify outliers or obvious cases of intrusion based on proximity to known patterns.
- **Cons**: The computational cost of KNN increases with the size of the dataset because it requires calculating distances to all other points. It is less suitable for real-time applications with very large datasets.
- **Motivation**: KNN was selected as an initial filtering tool due to its simplicity and ability to quickly identify obvious patterns of benign or malicious behavior. However, due to its compu- tational demands, it is best used for smaller datasets or in scenarios where rapid decisions are needed without requiring high precision.

Weighted K-Nearest Neighbors (W-KNN) (Dudani, 1976):

- **Explanation**: W-KNN extends KNN by assigning weights to neighbors based on their distance, giving more influence to closer neighbors. This refinement helps improve accuracy, particularly in imbalanced datasets.
- **Pros**: Provides better accuracy compared to standard KNN, especially in handling imbalanced classes, which is common in intrusion detection datasets.
- **Cons**: Similar to KNN, W-KNN is computationally intensive and may struggle with real-time requirements when applied to large datasets.
- **Motivation**: W-KNN was chosen for its enhanced accuracy over KNN in detecting intrusions in imbalanced datasets. It offers a more refined approach to preliminary filtering, making it suitable for identifying cases where there is a subtle distinction between benign and malicious packets.

Random Forest Classifier (RFC) (Breiman, 2001):

- **Explanation**: RFC is an ensemble learning method that builds multiple Decision Trees and com- bines their predictions. It is highly effective in capturing complex interactions between features.
- **Pros**: Offers high accuracy and robustness, particularly in datasets with varied and complex patterns. It also handles large datasets well and is less prone to overfitting due to its ensemble nature.
- **Cons**: RFC can be computationally expensive, particularly in the training phase, and may not be the fastest option for real-time filtering.
- **Motivation**: RFC was included for its ability to handle complex patterns in the data, making it a strong candidate for initial filtering in scenarios where precision is critical. While more resource-intensive than KNN, its high accuracy makes it valuable for filtering out packets that exhibit more subtle signs of malicious behavior.

Multilayer Perceptron (MLP) (Rumelhart et al., 1986):

- **Explanation**: MLP is a type of artificial neural network with multiple layers that can model non-linear relationships in the data.
- **Pros**: Capable of capturing complex patterns and non-linear relationships, making it suitable for scenarios where malicious activities do not follow straightforward patterns.
- **Cons**: Requires significant computational resources and careful tuning to avoid overfitting. It is less efficient for real-time filtering due to its complexity.
- **Motivation**: MLP was included for its ability to model complex, non-linear interactions that might be indicative of sophisticated intrusion attempts. However, due to its resource-intensive nature, it is best applied after simpler models have filtered out more obvious cases.

Decision Tree Classifier (DTC) (Quinlan, 1986):

- **Explanation**: DTC predicts the value of a target variable by learning simple decision rules inferred from the data features. It builds a model in the form of a tree structure, where nodes represent features and branches represent decision rules.
- **Pros**: Simple, easy to interpret, and quick to execute, making it ideal for initial, rapid filtering of network traffic.
- **Cons**: Prone to overfitting, especially in deep trees, and can be unstable if the dataset is diverse.
- **Motivation**: DTC was chosen for its simplicity and speed, making it an excellent candidate for initial filtering. Its ability to make quick decisions based on straightforward rules allows for the efficient classification of clear-cut cases, reducing the load on more complex models.

Deep Learning (DL) (LeCun et al., 2015):

- **Explanation**: Deep Learning models, typically comprising multiple layers of neurons, are ca- pable of learning from vast amounts of data. They excel at discovering complex patterns and representations in the data.
- **Our Model**: We used a Sequential model from the Keras library (Chollet et al., 2015), featuring multiple dense layers with ReLU and Sigmoid activations, and dropout for regularization. This model was designed to explore the deep patterns in the network traffic data.
- **Pros**: Exceptionally powerful in detecting complex patterns, with high potential accuracy in identifying sophisticated intrusion attempts.
- **Cons**: Extremely resource-intensive, both in terms of computation and memory. DL models also require large amounts of labeled data and are prone to overfitting without careful regularization.
- **Motivation**: While DL models offer high accuracy and the ability to model complex data pat- terns, they are considered overkill for the purposes of preliminary filtering. Nonetheless, we included DL in our study to benchmark its performance against simpler models and to highlight the trade-offs in using such a powerful tool for initial detection.

Convolutional Neural Networks (CNN):

- **Explanation**: CNNs are specialized deep learning models often used in image processing but can be adapted for other types of data with spatial hierarchies. They are known for their ability to automatically and efficiently capture complex patterns.
- **Pros**: Capable of learning detailed spatial features, making them highly accurate for tasks in- volving hierarchical data.
- **Cons**: Computationally heavy and requires significant resources to train and deploy. CNNs are also generally unnecessary for the type of one-dimensional data used in this study, making them less practical for preliminary network intrusion detection.
- **Motivation**: CNNs are included in our experiments as a demonstration of the upper limits of complexity and accuracy. However,

due to their resource demands, they are not recommended for the preliminary filtering stage. This model serves more as a contrast to simpler, faster methods, reinforcing our focus on efficiency and practicality.

5. Results and discussion

5.1. CICIoT2023

Tables 1 and 2 summarize the performance metrics, for the CICIoT2023 dataset, for each model when using only the IAT feature and the entire dataset, respectively, highlighting the trade-offs between compu- tational efficiency and accuracy.

Fig. 3 shows a comparison between recall scores and computational times for different models using the IAT-only feature and the whole dataset. The recall scores are consistently high across both methods, while the IAT-only approach significantly reduces training and prediction times, highlighting its efficiency for real-time detection.

5.1.1. K-Nearest neighbors (KNN)

The K-Nearest Neighbors (KNN) model demonstrated robust performance in both the IAT-only and the whole dataset scenarios. For the IAT-only feature, KNN achieved an accuracy of 99.48 %, a recall of 99.07 %, an F1 score of 99.49 %, and a precision of 99.92 %. The ROC AUC score was 99.60 %, indicating a strong ability to discriminate between classes. The False Positive Rate (FPR) was notably low at 0.08 %, underscoring the model's reliability in minimizing false alarms.

When utilizing the entire dataset, KNN's accuracy slightly decreased to 99.42 %, and its recall dropped to 98.85 %, which is still very high but indicates a marginal performance decline compared to the IAT-only approach. The F1 score also slightly dropped to 99.43 %, and the precision remained very high at 99.98 %. The ROC AUC remained stable at 99.60 %, while the FPR decreased even further to 0.02 %. Although the performance metrics are quite similar, the prediction time for the whole dataset approach increased significantly from 0.0074 s (IAT-only) to 0.1983 s (whole dataset), highlighting the efficiency of the IAT-only approach for real-time detection. It's important to note that training times are irrelevant here since KNN doesn't involve a traditional training algorithm; hence, the focus is on prediction times.

The trade-off analysis highlights the efficiency of the IAT-only approach, especially in scenarios where real-time processing is critical. While there is a slight decrease in recall and accuracy, the reduction in pre- diction time makes the IAT-only approach particularly appealing. The negligible differences in performance metrics suggest that the IAT-only feature is sufficient for high-accuracy detection, especially when the speed of detection is a priority. Therefore, for applications where rapid response is essential, the IAT-only approach provides an excellent balance between speed and accuracy.

5.1.2. Weighted K-Nearest neighbors (W-KNN)

Weighted K-Nearest Neighbors (W-KNN) performed similarly to KNN, with slight variations. Using the IAT-only feature, W-KNN achieved an accuracy of 99.35 %, a recall of 99.13 %, an F1 score of 99.35 %, and a precision of 99.56 %. The ROC AUC was 99.59 %, and the FPR was 0.42 %, which, although slightly higher than KNN, still indicates strong performance.

With the whole dataset, W-KNN's accuracy marginally increased to 99.43 %, and the recall slightly decreased to 98.88 %. The F1 score was consistent at 99.42 %, and the precision increased to 99.99 %, reflecting a near-perfect classification ability. The ROC AUC improved slightly to 99.61 %, and the FPR dropped to 0.01 %, indicating better performance in terms of minimizing false positives.

The trade-off analysis for W-KNN reveals that the IAT-only approach offers significant benefits in terms of prediction speed without compromising much on accuracy. While the training time for IAT-only was 0.9863 s, slightly higher than the whole dataset's 0.9827 s, the prediction time was much faster at 0.0054 s compared to 0.2265 s for the whole dataset. However, similar to KNN, the training time is largely irrelevant here, as W-KNN does not utilize a traditional training algorithm. The emphasis is therefore on prediction times, where the IAT-only approach offers a substantial advantage. This drastic improvement in prediction time, coupled with the minimal impact on performance metrics, makes the IAT- only approach highly advantageous for real-time applications. The slight reduction in accuracy and recall is offset by the substantial gains in computational efficiency, making the IAT-only approach a strong candidate for scenarios where quick decision-making is essential.

5.1.3. Random forest classifier (RFC)

The Random Forest Classifier (RFC) provided strong results across both datasets, achieving a recall of 99.08 % with the IAT-only feature and 99.55 % with the whole dataset. The accuracy was also high, at 99.39 % for IAT-only and 99.76 % for the whole dataset. While the whole dataset approach slightly outperformed the IAT-only approach in recall and accuracy, the differences are minimal, indicating that IAT-only is a viable option for effective threat detection.

In terms of F1 score, the RFC model achieved 99.39 % with IAT-only and 99.77 % with the whole dataset, reflecting consistent performance. The precision was 99.70 % with IAT-only, slightly lower than the 99.99 % achieved with the whole dataset. The ROC AUC scores were also very close, with 99.68 % for IAT-only and 99.98 % for the whole dataset, indicating strong discriminative power in both cases. The FPR was 0.29 % for IAT-only and decreased to 0.01 % for the whole dataset, highlighting a minor improvement in reducing false positives with the full dataset.

The trade-off analysis for RFC indicates that while the whole dataset approach offers slightly better performance metrics, the IAT-only approach significantly reduces computational times. The training time for IAT-only was 295.4321 s, considerably lower than the 1462.3333 s required for the whole dataset. Similarly, the prediction time was reduced from 0.0156 s (whole dataset) to 0.0115 s (IAT-only). These improvements in speed, combined with only a slight reduction in performance, suggest that the IAT-only approach is a viable and efficient option, particularly in scenarios where computational resources are limited or quick responses are necessary.

Table 1
Averaged Performance Metrics over 30 Runs for "IAT" Feature Only: CICIoT2023.

Model	Accuracy (%)	F1 Score (%)	Recall (%)	Precision (%)	ROC AUC (%)	FPR (%)	Training Time (s)	Prediction Time (s)
KNN	99.48 %	99.49 %	99.07 %	99.92 %	99.60 %	0.08 %	0.8504	0.0074
W-KNN	99.35 %	99.35 %	99.13 %	99.56 %	99.59 %	0.42 %	0.9863	0.0054
RFC	99.39 %	99.39 %	99.08 %	99.70 %	99.68 %	0.29 %	295.4321	0.0115
MLP	74.12 %	79.02 %	98.98 %	66.00 %	74.10 %	50.25 %	576.7312	0.0021
DTC	99.39 %	99.39 %	99.07 %	99.71 %	99.50 %	0.28 %	3.9135	0.0013
DL	74.18 %	79.22 %	98.99 %	66.35 %	74.20 %	49.80 %	765.3311	0.1102
CNN	74.22 %	79.20 %	99.01 %	66.32 %	74.25 %	49.85 %	1210.7654	0.1756

Table 2
Averaged Performance Metrics over 30 Runs for Whole Dataset: CICIoT2023.

Model	Accuracy (%)	F1 Score (%)	Recall (%)	Precision (%)	ROC AUC (%)	FPR (%)	Training Time (s)	Prediction Time (s)
KNN	99.42 %	99.43 %	98.85 %	99.98 %	99.60 %	0.02 %	0.8958	0.1983
W-KNN	99.43 %	99.42 %	98.88 %	99.99 %	99.61 %	0.01 %	0.9827	0.2265
RFC	99.76 %	99.77 %	99.55 %	99.99 %	99.98 %	0.01 %	1462.3333	0.0156
MLP	99.42 %	99.43 %	98.92 %	99.93 %	99.81 %	0.03 %	967.5534	0.0031
DTC	99.71 %	99.71 %	99.65 %	99.75 %	99.72 %	0.18 %	150.6756	0.0021
DL	99.35 %	99.36 %	98.80 %	99.95 %	99.75 %	0.05 %	781.4123	0.1267
CNN	99.38 %	99.39 %	98.81 %	99.98 %	99.78 %	0.00 %	3665.7314	0.2593

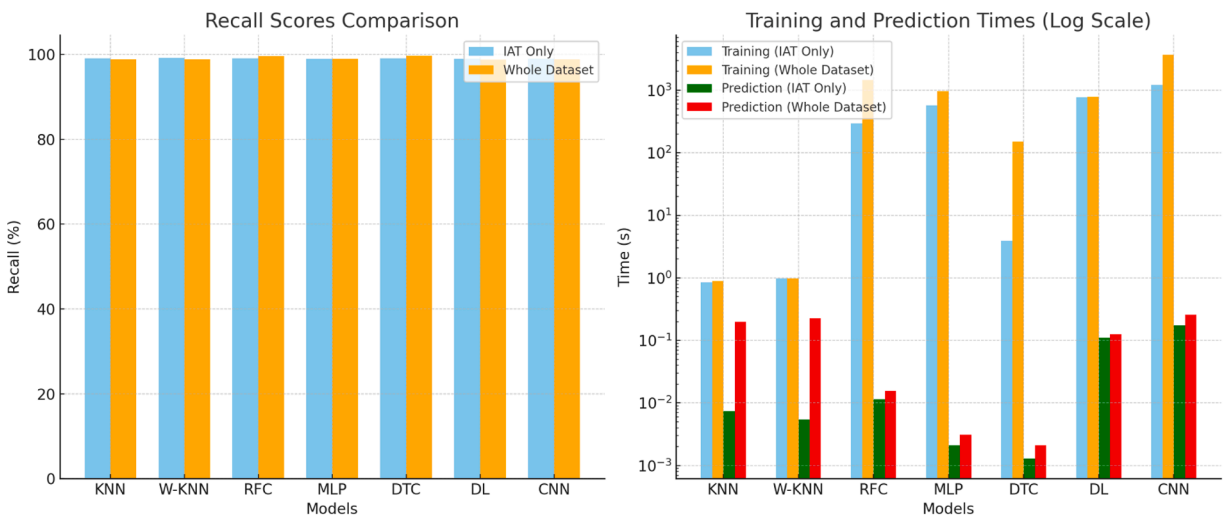

Fig. 3. Recall scores and computational times comparison for IAT-only vs. whole dataset across various models for CI- CIoT2023 dataset.

5.1.4. Multilayer perceptron (MLP)

The Multilayer Perceptron (MLP) showed significant differences in performance between the IAT-only and whole dataset approaches. With the IAT-only feature, MLP achieved an accuracy of 74.12 %, a recall of 98.98 %, an F1 score of 79.02 %, and a precision of 66.00 %. The ROC AUC score was 74.10 %, and the FPR was relatively high at 50.25 %, indicating a higher rate of false positives compared to other models.

Using the whole dataset, the MLP's performance improved significantly. The accuracy jumped to 99.42 %, the recall to 98.92 %, the F1 score to 99.43 %, and the precision to 99.93 %. The ROC AUC score increased substantially to 99.81 %, and the FPR dropped to just 0.03 %. These improvements demonstrate the MLP's ability to leverage the additional information in the whole dataset for better performance.

The trade-off analysis for MLP reveals that while the whole dataset approach yields better performance metrics, the IAT-only approach still provides substantial recall (98.98 %) with much lower computational demands. The training time for the IAT-only feature was 576.7312 s, significantly lower than the 967.5534 s required for the whole dataset. The prediction time was also lower for IAT-only at 0.0021 s compared to 0.0031 s for the whole dataset. Given these trade-offs, the IAT-only approach may be preferable in environments where computational resources are constrained, or where the primary goal is to maintain a high recall with minimal processing time.

5.1.5. Decision tree classifier (DTC)

The Decision Tree Classifier (DTC) performed well in both scenarios. With the IAT-only feature, DTC achieved an accuracy of 99.39 %, a recall of 99.07 %, an F1 score of 99.39 %, and a precision of 99.71 %. The ROC AUC score was 99.50 %, and the FPR was low at 0.28 %.

For the whole dataset, DTC's performance metrics improved slightly. The accuracy increased to 99.71 %, the recall to 99.65 %, the F1 score to 99.71 %, and the precision to 99.75 %. The ROC AUC score was 99.72 %, and the FPR decreased to 0.18 %. These results indicate that while the whole dataset provides marginal improvements, the IAT-only approach still offers high accuracy and recall.

The trade-off analysis for DTC shows that the IAT-only approach provides a good balance between accuracy and computational efficiency. The training time for the IAT-only feature was 3.9135 s, significantly lower than the 150.6756 s required for the whole dataset. The prediction time was also faster for IAT-only at 0.0013 s compared to 0.0021 s for the whole dataset. Given these factors, the IAT-only approach is highly effective in scenarios where quick decision-making is critical, offering high accuracy with reduced computational overhead.

5.1.6. Deep learning (DL)

The Deep Learning (DL) model showed considerable differences in performance between the two ap- proaches. With the IAT-only feature, DL achieved an accuracy of 74.18 %, a recall of 98.99 %, an F1 score of 79.22 %, and a precision of 66.35 %. The ROC AUC score was 74.20 %, and the FPR was 49.80 %, indicating a higher rate of false positives.

Using the whole dataset, DL's performance improved significantly. The accuracy increased to 99.35 %, the recall to 98.80 %, the F1 score to 99.36 %, and the precision to 99.95 %. The ROC AUC score also saw a substantial increase to 99.75 %, and the FPR dropped dramatically to just 0.05 %. These results highlight the DL model's ability to extract more meaningful patterns from the full dataset, leading to better overall performance.

The trade-off analysis for the DL model underscores the significant computational benefits of using the IAT-only approach. The training time for the IAT-only feature was 765.3311 s, which, although substantial, is still less than the 781.4123 s required for the whole dataset. The prediction time for IAT-only was 0.1102 s, compared to 0.1267 s for

the whole dataset. While the whole dataset offers better performance metrics, the IAT-only approach still provides a high recall with faster processing times, making it suitable for scenarios where quick decision-making is essential, and computational resources are limited.

5.1.7. Convolutional neural networks (CNN)

Convolutional Neural Networks (CNN) demonstrated similar trends as other models, with significant performance improvements when using the whole dataset. For the IAT-only feature, CNN achieved an accuracy of 74.22 %, a recall of 99.01 %, an F1 score of 79.20 %, and a precision of 66.32 %. The ROC AUC score was 74.25 %, and the FPR was 49.85 %, reflecting a relatively high rate of false positives.

When using the whole dataset, CNN's performance metrics improved markedly. The accuracy rose to 99.38 %, the recall to 98.81 %, the F1 score to 99.39 %, and the precision to 99.98 %. The ROC AUC score increased significantly to 99.78 %, and the FPR dropped to nearly zero at 0.00 %. These results indicate that CNN benefits greatly from the additional features provided by the whole dataset.

The trade-off analysis for CNN highlights the substantial computational savings offered by the IAT- only approach. The training time for IAT-only was 1210.7654 s, much lower than the 3665.7314 s required for the whole dataset. The prediction time also improved, with IAT-only taking 0.1756 s compared to 0.2593 s for the whole dataset. Despite the superior performance metrics of the whole dataset, the IAT-only approach offers a highly efficient alternative, particularly in environments where computational resources are constrained, or where rapid inference is necessary.

5.1.8. Trade-Off analysis and conclusion

Across all models, the IAT-only approach consistently demonstrated significant computational efficiency, with minimal impact on key performance metrics such as recall, accuracy, and F1 score. While the whole dataset generally offered slightly better performance, the differences were often marginal, particularly in recall, where the IAT-only approach still achieved high values. The reduction in training and prediction times with the IAT-only approach is particularly valuable for real-time applications where speed is critical.

The trade-off between computational efficiency and model performance suggests that the IAT-only ap- proach is highly advantageous in scenarios where rapid detection is required, and computational resources are limited. The IAT-only approach provides a practical and effective solution for intrusion detection sys- tems, offering a good balance between speed and accuracy without the need for extensive computational resources. Given these benefits, the IAT-only approach stands out as a viable and efficient alternative to using the entire dataset, particularly in real-time or resource-constrained environments.

5.2. UNSW-NB15

Tables 3 and 4 summarize the performance metrics, for the UNSW-NB15 dataset, for each model when using only the IAT feature and the entire dataset, respectively, highlighting the trade-offs between computational efficiency and accuracy.

Fig. 4 shows a comparison between recall scores and computational times for different models using the IAT-only feature and the whole dataset for the UNSW-NB15 dataset. The results demonstrate that while recall scores generally improve with the full dataset, the IAT-only approach significantly reduces computational time, making it a viable option for real-time applications.

5.2.1. K-Nearest neighbors (KNN)

The K-Nearest Neighbors (KNN) model exhibited strong performance metrics when using both the IAT- only feature and the whole dataset. For the IAT-only feature, KNN achieved an accuracy of 75.48 %, an F1 score of 77.35 %, and a recall of 83.25 %. The precision was slightly lower at 72.00 %, and the ROC AUC was 81.50 %. However, the false positive rate (FPR) was relatively high at 32.60 %, indicating a considerable number of false positives.

In contrast, when using the whole dataset, KNN's performance improved significantly across all metrics. The accuracy increased to 90.28 %, the F1 score to 90.05 %, and the recall to 87.10 %. Precision also improved to 93.15 %, and the ROC AUC rose to 96.60 %. The FPR decreased dramatically to 6.45 %, reflecting a much lower rate of false positives.

From a trade-off perspective, the IAT-only approach offers substantial computational benefits. The training time for the IAT-only feature was 0.1502 s, significantly higher than the 0.0195 s required for the whole dataset. However, training times are not particularly important here, as KNN does not involve a traditional training algorithm. The prediction time, which is more critical in KNN, was much faster for the IAT-only approach at 0.0041 s compared to 0.0201 s for the whole dataset. This highlights the efficiency of the IAT-only approach, making it a viable option when computational resources are limited, particularly for scenarios requiring rapid predictions.

5.2.2. Weighted K-Nearest neighbors (W-KNN)

The Weighted K-Nearest Neighbors (W-KNN) model showed similar trends to KNN, with notable per- formance improvements when using the whole dataset. For the IAT-only feature, W-KNN achieved an accuracy of 76.05 %, an F1 score of 78.02 %, and a recall of 84.75 %. Precision was 72.25 %, the ROC AUC was 81.65 %, and the FPR was 32.72 %, slightly higher than that of the KNN model.

Using the whole dataset, W-KNN's performance metrics improved significantly. The accuracy increased to 91.47 %, the F1 score to 91.18 %, and the recall to 87.90 %. Precision also saw a notable increase to 94.60 %, while the ROC AUC improved to 97.12 %. The FPR dropped to 5.00 %, indicating a substantial reduction in false positives.

The trade-off analysis for W-KNN mirrors that of KNN, with the IAT-only approach offering significant computational advantages. The training time for the IAT-only feature was 0.1023 s, higher than the 0.0187 s for the whole dataset. However, training times are not particularly important here, as W-KNN, like KNN, does not utilize a traditional training algorithm. The prediction time was faster for the IAT-only approach at 0.0035 s compared to 0.0203 s for the whole dataset. The IAT- only approach thus provides a faster, more efficient alternative for real-time applications, particularly in environments with limited

Table 3
Averaged Performance Metrics over 30 Runs for "IAT" Feature Only: UNSW-NB15.

Model	Accuracy (%)	F1 Score (%)	Recall (%)	Precision (%)	ROC AUC (%)	FPR (%)	Training Time (s)	Prediction Time (s)
KNN	75.48 %	77.35 %	83.25 %	72.00 %	81.50 %	32.60 %	0.1502	0.0041
W-KNN	76.05 %	78.02 %	84.75 %	72.25 %	81.65 %	32.72 %	0.1023	0.0035
RFC	74.95 %	77.00 %	83.85 %	71.20 %	81.10 %	34.15 %	29.8500	0.0097
MLP	52.12 %	67.40 %	98.75 %	51.15 %	65.72 %	94.65 %	29.8105	0.0018
DTC	74.67 %	76.55 %	82.45 %	71.28 %	80.32 %	33.32 %	0.3645	0.0012
DL	51.95 %	67.10 %	98.12 %	51.10 %	65.61 %	94.40 %	108.6215	0.1003
CNN	52.15 %	67.60 %	99.70 %	51.20 %	52.22 %	95.60 %	190.7123	0.2054

Table 4
Averaged Performance Metrics over 30 Runs for Whole Dataset: UNSW-NB15.

Model	Accuracy (%)	F1 Score (%)	Recall (%)	Precision (%)	ROC AUC (%)	FPR (%)	Training Time (s)	Prediction Time (s)
KNN	90.28 %	90.05 %	87.10 %	93.15 %	96.60 %	6.45 %	0.0195	0.0201
W-KNN	91.47 %	91.18 %	87.90 %	94.60 %	97.12 %	5.00 %	0.0187	0.0203
RFC	94.80 %	94.68 %	93.20 %	96.22 %	99.22 %	3.75 %	68.9510	0.0135
MLP	91.55 %	91.45 %	90.68 %	92.30 %	98.05 %	7.70 %	375.1425	0.0049
DTC	93.70 %	93.63 %	92.88 %	94.30 %	94.42 %	5.65 %	7.5340	0.0018
DL	89.65 %	89.48 %	87.80 %	91.15 %	97.20 %	8.50 %	130.4825	0.1352
CNN	91.75 %	91.58 %	89.30 %	93.90 %	98.18 %	5.82 %	341.8204	0.2121

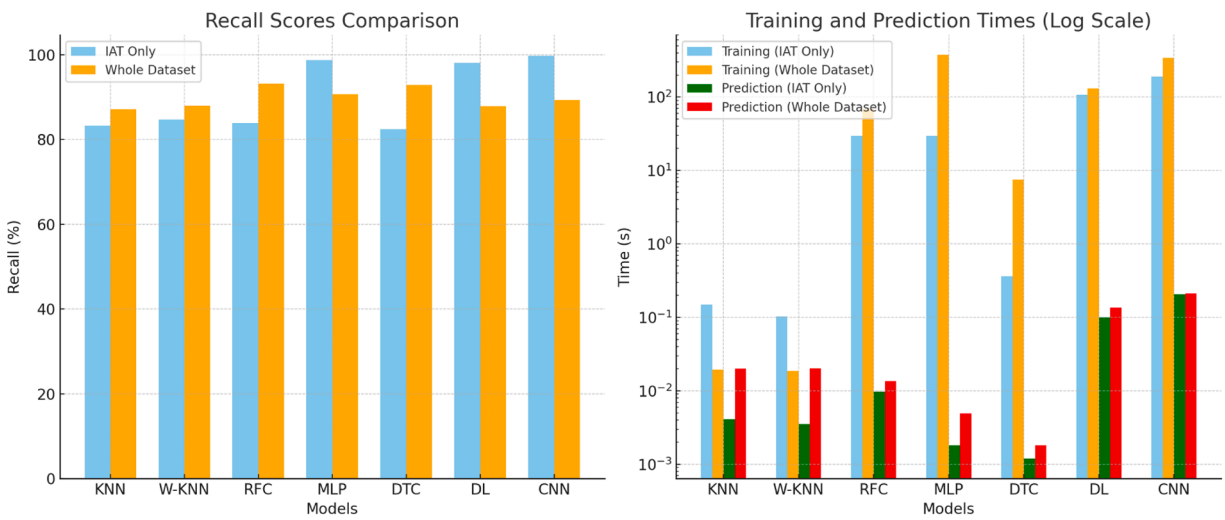

Fig. 4. Recall scores and computational times comparison for IAT-only vs. whole dataset across various models for UNSW- NB15 dataset.

computational resources.

5.2.3. *Random forest classifier (RFC)*

The Random Forest Classifier (RFC) demonstrated strong performance across both the IAT-only and whole dataset approaches. For the IAT-only feature, RFC achieved an accuracy of 74.95 %, an F1 score of 77.00 %, and a recall of 83.85 %. Precision was 71.20 %, the ROC AUC was 81.10 %, and the FPR was 34.15 %, indicating a moderate number of false positives.

With the whole dataset, RFC's performance improved significantly. The accuracy increased to 94.80 %, the F1 score to 94.68 %, and the recall to 93.20 %. Precision also improved to 96.22 %, and the ROC AUC rose to 99.22 %. The FPR decreased to 3.75 %, reflecting a much lower rate of false positives.

The trade-off analysis for RFC highlights the computational savings offered by the IAT-only approach. The training time for the IAT-only feature was 29.8500 s, compared to 68.9510 s for the whole dataset. The prediction time was also shorter for the IAT-only approach at 0.0097 s compared to 0.0135 s for the whole dataset. While the whole dataset offers better performance metrics, the IAT- only approach provides a more efficient alternative, particularly in scenarios where computational resources are limited or where rapid detection is crucial.

5.2.4. *Multilayer perceptron (MLP)*

The Multilayer Perceptron (MLP) model displayed contrasting results between the IAT-only feature and the whole dataset. For the IAT-only feature, MLP achieved an accuracy of 52.12 %, an F1 score of 67.40 %, and a very high recall of 98.75 %. However, the precision was low at 51.15 %, and the ROC AUC was 65.72 %. The FPR was significantly high at 94.65 %, indicating a very high number of false positives.

Using the whole dataset, MLP's performance metrics improved considerably. The accuracy increased to 91.55 %, the F1 score to 91.45 %, and the recall to 90.68 %. Precision also improved to 92.30 %, while the ROC AUC rose to 98.05 %. The FPR decreased to 7.70 %, indicating a much lower rate of false positives.

The trade-off analysis for MLP suggests that while the whole dataset offers better overall performance, the IAT-only approach provides significant computational savings. The training time for the IAT-only feature was 29.8105 s, compared to 375.1425 s for the whole dataset. The prediction time was also shorter for the IAT-only approach at 0.0018 s compared to 0.0049 s for the whole dataset. Despite the performance improvements with the whole dataset, the IAT-only approach still provides a high recall with faster processing times, making it suitable for scenarios where rapid detection is essential, and computational resources are constrained.

5.2.5. *Decision tree classifier (DTC)*

The Decision Tree Classifier (DTC) exhibited robust performance with both the IAT-only feature and the whole dataset. For the IAT-only feature, DTC achieved an accuracy of 74.67 %, an F1 score of 76.55 %, and a recall of 82.45 %. Precision was 71.28 %, the ROC AUC was 80.32 %, and the FPR was 33.32 %, indicating a moderate number of false positives.

Using the whole dataset, DTC's performance metrics improved substantially. The accuracy increased to 93.70 %, the F1 score to 93.63 %, and the recall to 92.88 %. Precision also improved to 94.30 %, while the ROC AUC rose to 94.42 %. The FPR decreased to 5.65 %, reflecting a much lower rate of false positives.

The trade-off analysis for DTC highlights the computational benefits of the IAT-only approach. The training time for the IAT-only feature was 0.3645 s, compared to 7.5340 s for the whole dataset. The prediction time was also shorter for the IAT-only approach at 0.0012 s compared to

0.0018 s for the whole dataset. While the whole dataset offers better performance metrics, the IAT-only approach provides a more efficient alternative, particularly in environments where computational resources are limited, or where rapid detection is crucial.

5.2.6. Deep learning (DL)

The Deep Learning (DL) model showed a significant contrast between the IAT-only feature and the whole dataset. For the IAT-only feature, DL achieved an accuracy of 51.95 %, an F1 score of 67.10 %, and a very high recall of 98.12 %. However, the precision was low at 51.10 %, and the ROC AUC was 65.61 %. The FPR was significantly high at 94.40 %, indicating a very high number of false positives.

With the whole dataset, DL's performance metrics improved markedly. The accuracy increased to 89.65 %, the F1 score to 89.48 %, and the recall to 87.80 %. Precision also improved to 91.15 %, while the ROC AUC rose to 97.20 %. The FPR decreased to 8.50 %, indicating a much lower rate of false positives.

The trade-off analysis for DL underscores the significant computational benefits of the IAT-only approach. The training time for the IAT-only feature was 108.6215 s, compared to 130.4825 s for the whole dataset. The prediction time was also shorter for the IAT-only approach at 0.1003 s compared to 0.1352 s for the whole dataset. While the whole dataset offers better performance metrics, the IAT- only approach still provides a high recall with faster processing times, making it suitable for scenarios where quick decision-making is essential, and computational resources are limited.

5.2.7. Convolutional neural networks (CNN)

Convolutional Neural Networks (CNN) exhibited similar trends to other models, with significant perfor- mance improvements when using the whole dataset. For the IAT-only feature, CNN achieved an accuracy of 52.15 %, an F1 score of 67.60 %, and a very high recall of 99.70 %. However, the precision was notably low at 51.20 %, and the ROC AUC was 52.22 %. The FPR was also quite high at 95.60 %, indicating a significant number of false positives.

Using the whole dataset, CNN's performance metrics improved across the board. The accuracy increased to 91.75 %, the F1 score to 91.58 %, and the recall to 89.30 %. Precision also saw a substantial increase to 93.90 %, and the ROC AUC rose dramatically to 98.18 %. The FPR decreased to 5.82 %, reflecting a significant reduction in false positives.

The trade-off analysis for CNN highlights the substantial computational savings offered by the IAT- only approach. The training time for the IAT-only feature was 190.7123 s, compared to 341.8204 s for the whole dataset. The prediction time was also shorter for the IAT-only approach at 0.2054 s compared to 0.2121 s for the whole dataset. While the whole dataset provides better overall performance metrics, the IAT-only approach still delivers a very high recall and faster processing times, making it particularly useful in real-time detection scenarios where computational efficiency is paramount.

5.2.8. Trade-Off analysis and conclusion

Overall, the analysis of the UNSW-NB15 dataset using various models highlights the potential of the IAT-only approach, particularly in contexts where computational efficiency and rapid detection are critical. The IAT-only approach consistently offers reduced training and prediction times across all models, which is a significant advantage in environments where processing speed is crucial.

While the whole dataset generally provides better accuracy, F1 scores, precision, and lower false positive rates, the IAT-only approach still maintains competitive recall scores, particularly in models like CNN, DL, and MLP. This suggests that the IAT-only approach can be a viable option in scenarios where high recall is prioritized over other metrics, such as in preliminary filtering stages of intrusion detection systems.

In summary, the IAT-only approach offers a compelling balance between computational efficiency and detection performance. It is especially beneficial in situations where rapid, real-time decisions are required, and resources are limited. Although it may sacrifice some accuracy and precision, its ability to maintain high recall with significantly faster processing times makes it a strong candidate for initial threat detection layers, where speed and efficiency are critical.

5.3. CIC-IDS-2017

Tables 5 and 6 summarize the performance metrics, for the CIC-IDS-2017 dataset, for each model when using only the IAT feature and the entire dataset, respectively, highlighting the trade-offs between computational efficiency and accuracy.

Fig. 5 presents a comparison of recall scores and computational times for various models using the IAT-only feature versus the entire dataset for the CIC-IDS-2017 dataset. The results indicate that although recall scores tend to be higher when using the full dataset, the IAT-only approach considerably lowers computational time, making it a practical choice for real-time applications.

5.3.1. K-Nearest neighbors (KNN)

For the CIC-IDS-2017 dataset, the KNN model utilizing only the "IAT" feature achieved an accuracy of 73.14 %, with an F1 score of 72.11 % and a recall of 69.84 %. Precision was recorded at 74.92 %, while the ROC AUC reached 81.18 %. However, the false positive rate (FPR) was relatively high at 22.76 %. In terms of computational efficiency, the KNN model exhibited a training time of 0.9123 s and a prediction time of 0.0041 s, indicating that this approach is computationally light. However, it's important to note that training times are not particularly relevant here, as KNN does not involve a traditional training algorithm.

When the entire dataset was used, KNN's performance metrics improved significantly. The accuracy jumped to 99.27 %, with an F1 score of 99.26 % and a recall of 99.71 %. Precision also improved substantially to 98.84 %, with the ROC AUC increasing to 99.74 %. The FPR was drastically reduced to 1.22 %. However, these performance gains were accompanied by an increase in training time to 0.7923 s and prediction time to 0.3629 s.

The trade-off for KNN between using only the IAT feature and the entire dataset reveals a significant decision point depending on the application. The IAT-only approach offers significantly reduced computa- tional times, particularly in prediction, which is crucial in real-time detection scenarios. However, this speed comes at the cost of accuracy, precision, and FPR. The whole dataset approach, while slightly more com- putationally intensive, offers far superior performance metrics, making it the better choice for environments where accuracy is paramount. However, in time-sensitive situations, the IAT-only approach could still be valuable, especially given its much shorter prediction time of 0.0041 s compared to 0.3629 s with the full dataset.

5.3.2. Weighted K-Nearest neighbors (W-KNN)

The W-KNN model, much like KNN, showed moderate performance when using the "IAT" feature alone. The model achieved an accuracy of 72.83 %, with an F1 score of 71.98 %, recall of 70.02 %, and precision of 74.48 %. The ROC AUC was slightly lower at 80.81 %, and the FPR stood at 23.47 %. Computational times were 0.5027 s for training and 0.0028 s for prediction, making it efficient in terms of computation. However, it's important to note that training times are not particularly relevant here, as W-KNN, like KNN, does not utilize a traditional training algorithm.

Using the entire dataset, W-KNN's performance improved significantly. The accuracy increased to 99.35 %, with an F1 score of 99.36 %, and recall of 99.76 %. Precision improved to 98.93 %, and the ROC AUC reached 99.74 %, while the FPR decreased to 1.09 %. Training and prediction times increased to 1.4792 s and 0.1538 s, respectively.

The analysis of W-KNN demonstrates that while the IAT-only approach is advantageous in terms of speed, it sacrifices some

Table 5
Averaged Performance Metrics over 30 Runs for "IAT" Feature Only: CIC-IDS-2017.

Model	Accuracy (%)	F1 Score (%)	Recall (%)	Precision (%)	ROC AUC (%)	FPR (%)	Training Time (s)	Prediction Time (s)
KNN	73.14 %	72.11 %	69.84 %	74.92 %	81.18 %	22.76 %	0.9123	0.0041
W-KNN	72.83 %	71.98 %	70.02 %	74.48 %	80.81 %	23.47 %	0.5027	0.0028
RFC	66.13 %	54.46 %	41.22 %	82.09 %	72.05 %	9.28 %	158.9642	0.0061
MLP	64.47 %	46.12 %	31.15 %	94.05 %	71.88 %	2.03 %	405.3423	0.0028
DTC	65.98 %	53.98 %	40.09 %	83.02 %	70.14 %	8.73 %	1.8342	0.0016
DL	64.95 %	46.43 %	31.25 %	95.28 %	66.54 %	1.62 %	350.2108	0.1108
CNN	65.12 %	46.15 %	31.08 %	95.62 %	65.76 %	1.52 %	425.3198	0.1894

Table 6
Averaged Performance Metrics over 30 Runs for Whole Dataset: CIC-IDS-2017.

Model	Accuracy (%)	F1 Score (%)	Recall (%)	Precision (%)	ROC AUC (%)	FPR (%)	Training Time (s)	Prediction Time (s)
KNN	99.27 %	99.26 %	99.71 %	98.84 %	99.74 %	1.22 %	0.7923	0.3629
W-KNN	99.35 %	99.36 %	99.76 %	98.93 %	99.74 %	1.09 %	1.4792	0.1538
RFC	99.88 %	99.88 %	99.93 %	99.84 %	99.98 %	0.16 %	232.5421	0.0103
MLP	98.63 %	98.64 %	99.78 %	97.52 %	99.82 %	2.58 %	3256.3174	0.0063
DTC	99.88 %	99.88 %	99.89 %	99.87 %	99.89 %	0.14 %	38.9427	0.0023
DL	98.57 %	98.59 %	99.64 %	97.46 %	99.79 %	2.58 %	371.2841	0.1207
CNN	99.37 %	99.38 %	99.23 %	99.48 %	99.95 %	0.52 %	2423.6843	0.2015

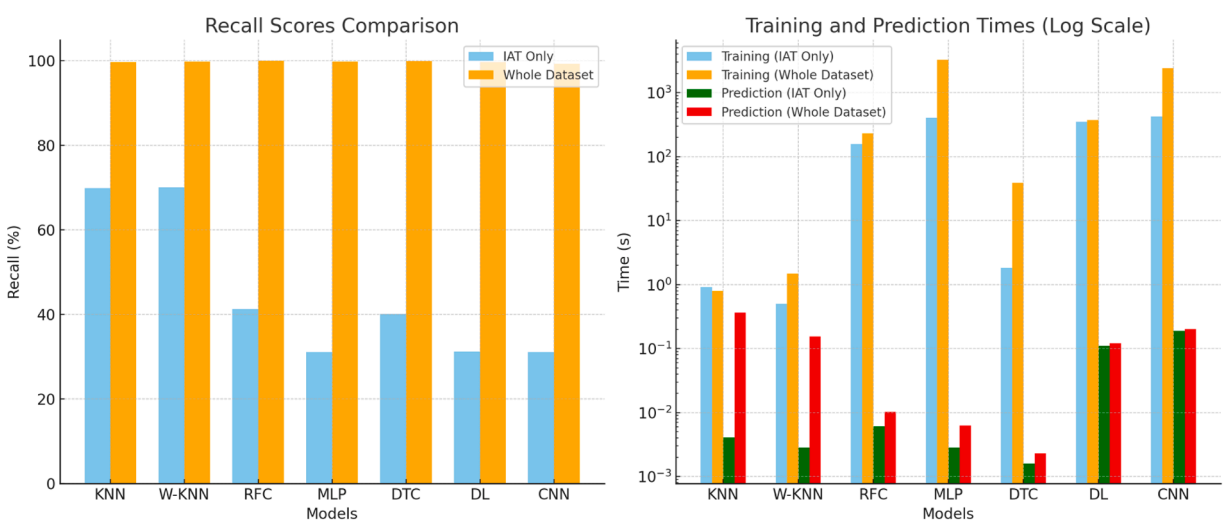

Fig. 5. Recall scores and computational times comparison for IAT-only vs. whole dataset across various models for CIC- IDS-2017 dataset.

accuracy, precision, and recall. The whole dataset approach, though requiring longer computational times, significantly enhances performance metrics, especially in recall and FPR. For scenarios where accuracy and reliability are crucial, the full dataset is preferable, but the IAT-only approach remains viable for real-time applications with its much faster prediction time of 0.0028 s compared to 0.1538 s with the whole dataset.

5.3.3. Random forest classifier (RFC)

For RFC using only the "IAT" feature, the model exhibited an accuracy of 66.13 %, with an F1 score of 54.46 % and a recall of 41.22 %. Precision was notably higher at 82.09 %, with a ROC AUC of 72.05 %. The FPR was relatively low at 9.28 %. However, the training time for this model was considerably higher at 158.9642 s, with a prediction time of 0.0061 s.

When applied to the entire dataset, RFC's performance metrics improved dramatically. The accuracy increased to 99.88 %, with an F1 score of 99.88 %, and recall reached 99.93 %. Precision improved to 99.84 %, with the ROC AUC reaching 99.98 %, and the FPR dropped to just 0.16 %. The increased performance, however, came with a substantial rise in computational resources, with training time extending to 232.5421 s and prediction time at 0.0103 s.

The trade-off analysis for RFC highlights that the IAT-only approach provides faster prediction times and lower computational demands, but at a considerable cost to accuracy, recall, and FPR. The whole dataset approach, though computationally more expensive, offers significantly better performance across all metrics, making it the preferred choice for applications where high precision and low FPR are critical. However, in scenarios where rapid prediction is more important, the IAT-only method, with its faster prediction time of 0.0061 s compared to 0.0103 s, may be beneficial.

5.3.4. Multi-Layer perceptron (MLP)

Using the "IAT" feature only, the MLP model achieved an accuracy of 64.47 %, with an F1 score of 46.12 % and a recall of 31.15 %. Precision was very high at 94.05 %, while the ROC AUC was 71.88 %. The FPR was

extremely low at 2.03 %. However, the model had a substantial training time of 405.3423 s and a prediction time of 0.0028 s.

With the whole dataset, MLP's performance improved dramatically. The accuracy increased to 98.63 %, the F1 score to 98.64 %, and the recall to 99.78 %. Precision remained high at 97.52 %, and the ROC AUC reached 99.82 %. The FPR increased slightly to 2.58 %. However, these gains came with significantly longer training times at 3256.3174 s and prediction time at 0.0063 s.

For MLP, the IAT-only approach offers shorter prediction times and maintains a low FPR, but it sig- nificantly underperforms in recall and F1 score. The whole dataset approach, although much more com- putationally intensive, provides vastly superior performance metrics, making it the preferred option where accuracy is critical. However, for real-time systems with limited computational resources, the IAT-only approach, with its faster prediction time of 0.0028 s compared to 0.0063 s, may still hold value.

5.3.5. Decision tree classifier (DTC)

Using the "IAT" feature only, the DTC model achieved an accuracy of 65.98 %, with an F1 score of 53.98 % and a recall of 40.09 %. Precision was relatively high at 83.02 %, with a ROC AUC of 70.14 %. The FPR was 8.73 %. The model had a very short training time of 1.8342 s, with a prediction time of 0.0016 s.

When the entire dataset was used, DTC's performance metrics improved significantly. Accuracy increased to 99.88 %, the F1 score to 99.88 %, and recall to 99.89 %. Precision improved to 99.87 %, and the ROC AUC reached 99.89 %. The FPR dropped to 0.14 %. Training time increased to 38.9427 s, and prediction time to 0.0023 s.

The DTC model's trade-off analysis shows that while the IAT-only approach provides faster training and prediction times, it comes at the expense of accuracy, recall, and F1 scores. The whole dataset approach, though requiring more computational resources, delivers better performance across all metrics, making it the better choice for applications where accuracy is paramount. However, for applications where speed is critical, the IAT-only approach, with its prediction time of 0.0016 s compared to 0.0023 s, offers a viable alternative.

5.3.6. Deep learning (DL)

For the DL model, using the "IAT" feature only resulted in an accuracy of 64.95 %, with an F1 score of 46.43 % and a recall of 31.25 %. Precision was very high at 95.28 %, while the ROC AUC was 66.54 %. The FPR was extremely low at 1.62 %. The model had a substantial training time of 350.2108 s and a prediction time of 0.1108 s.

When using the whole dataset, DL's performance metrics improved dramatically. The accuracy increased to 98.57 %, the F1 score to 98.59 %, and the recall to 99.64 %. Precision remained high at 97.46 %, and the ROC AUC reached 99.79 %. The FPR increased slightly to 2.58 %. The training and prediction times were 371.2841 s and 0.1207 s, respectively.

The DL model's trade-off analysis demonstrates that while the IAT-only approach provides the advantage of shorter prediction times and a low FPR, it substantially underperforms in recall and F1 score. The whole dataset approach, although computationally more demanding, offers significantly better results in terms of overall detection accuracy and reliability, making it the preferred option when precision is crucial. However, for real-time applications where computational speed is critical, the IAT-only approach, with its prediction time of 0.1108 s compared to 0.1207 s with the full dataset, could be a viable alternative.

5.3.7. Convolutional neural network (CNN)

Using the "IAT" feature only, the CNN model achieved an accuracy of 65.12 %, with an F1 score of 46.15 % and a recall of 31.08 %. Precision was very high at 95.62 %, while the ROC AUC stood at 65.76 %. The FPR was extremely low at 1.52 %. However, the model required a significant amount of time for training, at 425.3198 s, with a prediction time of 0.1894 s.

When utilizing the whole dataset, CNN's performance metrics saw substantial improvements. Accuracy increased to 99.37 %, with an F1 score of 99.38 %, and recall rose to 99.23 %. Precision remained high at 99.48 %, and the ROC AUC reached 99.95 %. The FPR slightly increased to 0.52 %. Despite these performance gains, the computational demands also increased, with training time extending to 2423.6843 s and prediction time to 0.2015 s.

The CNN model's trade-off analysis indicates that while the IAT-only approach significantly reduces prediction time, it compromises on accuracy, recall, and F1 score. The whole dataset approach, though computationally expensive, delivers markedly better performance across all evaluation metrics, making it the preferred choice for scenarios where detection accuracy is of utmost importance. However, in contexts where time efficiency is a priority and slight reductions in accuracy are acceptable, the IAT-only approach, with its faster prediction time of 0.1894 s compared to 0.2015 s, may still be valuable.

5.3.8. Trade-Off analysis and conclusion

In summary, the choice between using the "IAT" feature only versus the whole dataset depends signif- icantly on the specific application requirements. The IAT-only approach consistently offers faster training and prediction times across all models, making it suitable for real-time detection scenarios where computa- tional resources are limited or where speed is a critical factor. However, this speed advantage often comes at the cost of lower accuracy, recall, F1 scores, and, in many cases, a higher FPR.

Conversely, the whole dataset approach, while more computationally intensive, provides superior per- formance in terms of accuracy, recall, precision, and F1 score. This makes it the preferable option for applications where the highest possible detection accuracy is essential, such as in highly sensitive security environments.

Therefore, the decision between these two approaches should be based on the operational demands of the specific intrusion detection system being deployed. The IAT-only method offers a practical solution for environments where computational efficiency is more critical than absolute precision, while the whole dataset approach is better suited for settings where detection performance cannot be compromised. The ability to choose between these methods allows for a flexible and tailored approach to intrusion detection, optimizing both speed and accuracy based on the needs of the application.

6. Critical analysis of IAT-Based detection: vulnerabilities and limitations

While the use of Interpacket Arrival Time (IAT) as a feature in intrusion detection systems (IDS) offers substantial benefits in terms of computational efficiency and speed, it is essential to recognize its inherent vulnerabilities. IAT is a low-level, time-based metric that can be exploited or manipulated by sophisticated attackers, posing significant challenges when employed as a standalone feature for threat detection. In this section, we provide a critical analysis of the potential avenues for adversarial manipulation and discuss the broader limitations of relying exclusively on IAT for intrusion detection.

6.1. Manipulation of IAT by adversaries

Although our IAT-based detection system demonstrates high accuracy under standard conditions, our current study does not include empirical experiments specifically designed to evaluate its performance against adversarial manipulations. While we acknowledge potential vulnerabilities—such as traffic shaping, timed packet injection, and other evasion techniques—the robustness of our approach under these conditions re- mains theoretical. Future work should incorporate adversarially modified datasets and controlled simulation experiments to quantitatively assess and enhance the system's resilience.

While Interpacket Arrival Time (IAT) provides valuable insights for

detecting abnormal traffic patterns, its reliance on timing data alone makes it susceptible to sophisticated adversarial manipulations. Attackers aware of an IAT-based detection system may exploit this knowledge to craft traffic that mimics legitimate network behavior, thus evading detection. Below, we explore several specific techniques that adversaries could use to manipulate IAT and undermine the robustness of an IAT-based intrusion detection system (IDS):

6.1.1. Traffic shaping and throttling

One of the most effective techniques adversaries can use is traffic shaping or throttling. This involves controlling the rate at which packets are transmitted to match the timing patterns typically observed in legitimate traffic. By adjusting the interpacket timing intervals to mirror the legitimate traffic of the target network, attackers can align their traffic with the normal baseline, thus evading detection. This manipulation could be particularly successful in networks where traffic patterns follow predictable rhythms, such as corporate environments where user behavior and communication intervals are relatively uniform. In such cases, the IAT-based IDS may struggle to detect any deviations, resulting in false negatives as malicious traffic gets classified as normal.

6.1.2. Timed packet injection

In timed packet injection, attackers carefully synchronize their malicious packet transmissions with the network's regular traffic flow. By analyzing and replicating the timing of legitimate packets, adversaries can inject their traffic into the stream at intervals that resemble benign behavior. For instance, they could monitor the network to identify typical communication cycles and then inject their packets during periods of expected network activity. This method poses a significant challenge in networks where traffic patterns exhibit variability, as IAT-based detection systems may not easily differentiate between legitimate variations and carefully timed malicious traffic. Such attacks are especially dangerous in scenarios where attackers aim to blend into real-time traffic flows, like web browsing or real-time communication, making detection particularly challenging.

6.1.3. Packet padding and bursting

Packet padding and bursting represent additional strategies attackers might use to manipulate IAT and evade detection. In packet padding, adversaries introduce artificial delays between packet transmissions to stretch out the timing intervals, simulating slower or more controlled traffic behavior. Conversely, packet bursting involves sending packets in rapid succession, followed by periods of inactivity, to create the appearance of bursty but legitimate traffic, as often seen in multimedia applications like video streaming or VoIP. These manipulations allow attackers to disguise their activity by exploiting natural traffic variations in specific types of network applications. For example, real-time communication systems inherently generate bursty traffic, and attackers can leverage this to blend their malicious traffic into these expected patterns, reducing the likelihood of triggering IAT-based alarms.

6.1.4. Exploiting network congestion

Network congestion introduces inherent variability in packet transmission times, which attackers could exploit to obfuscate their activities. In congested environments, where packet delays, retransmissions, and dropped packets are common, malicious actors can time their traffic to coincide with periods of high con- gestion. By transmitting their packets during moments of legitimate network delay, attackers can hide their activity in the noise of fluctuating IAT values caused by congestion. Since IAT-based systems may interpret these irregular timings as a result of network conditions rather than malicious intent, adversaries can evade detection more easily. This method is particularly concerning in large-scale or highly dynamic networks, where congestion is frequent, making it difficult for an IDS to distinguish between benign delays and deliberate attempts to manipulate IAT.

6.1.5. Advanced persistent threats (APTs) and low-and-slow attacks

Advanced Persistent Threats (APTs) and low-and-slow attacks present another significant challenge to IAT-based systems. In these attack types, adversaries operate over extended periods, sending small amounts of traffic at a rate that closely mirrors the normal traffic flow. By operating in this subtle manner, attackers avoid triggering the detection thresholds typically set in IAT-based systems, which are often tuned to detect more abrupt or frequent timing anomalies. Over time, these attacks can lead to significant data exfiltration or system compromise without being flagged by the IDS. The ability of APTs to blend into the network's baseline traffic further complicates detection efforts, especially in networks with varying traffic patterns.

6.1.6. Manipulation through encryption and tunneling

Finally, attackers may leverage encryption and tunneling protocols to obscure their activities while still manipulating IAT. In encrypted tunnels, packet content is hidden, but timing patterns remain exposed. Adversaries can send their traffic through encrypted tunnels or VPNs to evade content-based inspection while carefully controlling the packet timing to mimic benign encrypted traffic. This technique is particularly effective in environments where encrypted traffic is common, as it allows attackers to blend their activities with legitimate encrypted communications, making it more difficult for an IAT-based IDS to detect anomalies based solely on timing characteristics. Moreover, this kind of manipulation can also complicate correlation with other network events, as encrypted packets may traverse different network paths, further obscuring malicious activities.

6.2. Conclusion on adversarial manipulation of IAT

The reliance on IAT for intrusion detection, while efficient, introduces vulnerabilities that skilled adver- saries can exploit. By manipulating traffic through techniques such as traffic shaping, timed packet injection, packet bursting, and exploiting network congestion, attackers can bypass IAT-based systems. Furthermore, advanced persistent threats and the use of encryption add layers of complexity, making detection even more challenging. These limitations underscore the importance of employing IAT as part of a broader, multi- layered defense strategy, rather than as a standalone solution. Future work should focus on enhancing the robustness of IAT-based systems, potentially through the integration of complementary features that mitigate these adversarial risks, ensuring a more comprehensive defense against evolving attack strategies.

6.3. Broader limitations of relying on IAT

In addition to the potential for adversarial manipulation, the IAT-only approach has inherent limitations when it comes to detecting a wide range of sophisticated attack patterns. These limitations are driven by the fact that IAT primarily captures timing-related anomalies, which may not always be indicative of malicious activity. Some broader limitations include:

6.3.1. Detection of stealthy attacks

Stealthy attacks, such as Advanced Persistent Threats (APTs), often unfold over long periods with minimal deviations in network behavior. These attacks are carefully crafted to avoid triggering alarms by mimicking normal traffic flows, and the timing between packets may remain within expected ranges. Because IAT is focused on packet timing, it may fail to capture the subtle anomalies associated with such attacks. In these cases, a more holistic approach that considers multiple features beyond IAT, such as payload analysis or behavior-based patterns, is necessary to detect the intrusion.

6.3.2. Low-Volume attacks

Low-volume attacks, such as data exfiltration over a long period, can

also bypass IAT-based detection systems. These attacks operate by transmitting small amounts of data at irregular intervals, often staying below the radar of timing-based anomaly detectors. Since the traffic volume is minimal and spread out over time, IAT values might remain well within the expected range, leading to false negatives. The challenge here is that such attacks do not create significant timing anomalies, and therefore, IAT alone is insufficient for detection.

6.3.3. Complex multi-stage attacks

Multi-stage attacks, where the attacker first gains access to the network and then later executes different stages of the attack (such as privilege escalation, lateral movement, and data exfiltration), are often difficult to detect using IAT alone. Each stage may involve different types of traffic patterns that do not necessarily exhibit anomalous timing. For example, during the reconnaissance phase, an attacker may scan the network at a slow, methodical pace, producing IAT values that closely resemble normal traffic. In later stages, when sensitive data is being exfiltrated, the attacker may again manipulate packet timings to evade detection.

6.3.4. Generalization across diverse network environments

Another significant limitation of IAT-based detection is its sensitivity to the specific network environment in which it is deployed. Networks with highly variable traffic patterns, such as those found in large enterprise settings or IoT networks, may experience wide fluctuations in IAT due to legitimate network activity. In these environments, distinguishing between benign fluctuations and genuine threats becomes increasingly difficult. This lack of generalization across different network environments limits the applicability of IAT as a universal detection feature, necessitating the incorporation of additional features or detection layers.

6.3.5. Experimental validation and future directions

While our current study focuses on demonstrating the efficacy of an IAT-based detection system under real-world conditions, we acknowledge that extensive experimental validation of adversarial manipulation re- mains an important area for further research. Specifically, future work should include controlled simulations and adversarial experiments that assess how techniques such as traffic shaping, timed packet injection, and packet bursting impact the detection performance. Such experiments would provide quantitative measures of resilience against deliberate manipulation of IAT and help refine countermeasures, such as adversarial training and hybrid detection models. We thus highlight the need for a dedicated experimental framework that can simulate various adversarial conditions and evaluate the efficacy of the proposed mitigation strate- gies. This effort would not only solidify the theoretical arguments presented in this paper but also offer practical insights into enhancing the robustness of IAT-based systems against evolving threat tactics.

6.4. Mitigating the limitations of IAT-Based detection

Given the identified vulnerabilities and limitations of relying solely on IAT, several strategies can be implemented to mitigate these risks:

6.4.1. Feature enrichment

To address the limitations of IAT, we propose augmenting the IAT feature with other complementary time-based metrics, such as packet size, flow duration, and connection rates. Feature fusion techniques can improve the robustness of the detection system by providing a more comprehensive view of network behavior, thereby reducing the likelihood of evasion through timing manipulation alone.

6.4.2. Hybrid detection models

Combining IAT-based anomaly detection with signature-based methods or machine learning models trained on a wider range of features can enhance detection capabilities. This hybrid approach allows the system to detect both known attack signatures and previously unseen anomalies, providing a more balanced and comprehensive defense.

6.4.3. Adaptive thresholding and context-aware detection

Implementing adaptive thresholding mechanisms, where the detection system dynamically adjusts its sensitivity based on real-time network conditions, can help reduce the impact of false positives in congested or variable networks. Context-aware detection methods that take into account the broader network context, such as the type of application generating the traffic or the time of day, can also improve the accuracy of IAT-based detection.

6.4.4. Adversarial resilience and robust training

To improve the resilience of IAT-based systems against adversarial manipulation, it is essential to incor- porate adversarial training techniques. By training the system on both normal and manipulated IAT data, we can enhance its ability to detect sophisticated evasion attempts. Additionally, integrating reinforcement learning techniques may enable the system to adapt to evolving attack strategies over time.

6.5. Conclusion

While the IAT-based approach offers notable advantages in terms of computational efficiency and real- time detection, it is not without its limitations. The susceptibility of IAT to manipulation, along with its inability to generalize across diverse attack types and network environments, underscores the need for a more nuanced approach to intrusion detection. As such, the IAT feature should be viewed as a component of a multi-layered defense strategy, rather than a standalone solution. By combining IAT with additional features and employing hybrid detection methods, the overall robustness of the IDS can be significantly improved, making it more resilient to sophisticated and evolving threats.

7. Privacy implications of IAT-Based intrusion detection

The use of Interpacket Arrival Time (IAT) as a key feature in intrusion detection systems offers con- siderable advantages in terms of computational efficiency and effectiveness in identifying specific network anomalies. However, it is important to critically examine the potential privacy implications associated with relying solely on IAT for detecting malicious behavior. Although IAT focuses on the timing between packets rather than packet content, there are several scenarios in which leveraging IAT might inadvertently expose sensitive information about network behaviors, raising concerns in privacy-sensitive environments.

One of the main privacy concerns is that IAT analysis could reveal insights into the structure of the network and its internal behaviors. For example, in highly regulated environments such as financial or healthcare institutions, where sensitive communications occur, the timing patterns between packets can expose details about network activity, including usage patterns, traffic loads, and communication intervals between different endpoints. Although this does not reveal the content of the data, the metadata provided by IAT could allow malicious actors or even third-party monitoring systems to infer sensitive operational details, such as the frequency of transactions or periods of heavy communication, which could be exploited in a targeted attack or leak information that organizations may want to keep confidential.

Another potential issue arises from the fact that IAT can expose behavioral fingerprints unique to individual users or systems. By analyzing the specific timing of packet exchanges, adversaries may be able to deduce patterns of system usage, user behavior, or even identify unique operational patterns that link activities to specific individuals. This can be particularly concerning in environments where anonymity and privacy are critical, such as government systems, secure corporate

networks, or whistleblowing platforms. The ability to track and profile users based on packet timing could undermine the privacy guarantees that organizations seek to provide.

Finally, the focus on IAT could also have implications when network traffic is encrypted. Even though the payload content of encrypted traffic is obfuscated, IAT values remain visible. This means that while attackers cannot access the data being transmitted, they can still analyze the timing intervals, potentially leading to indirect privacy breaches. For example, traffic patterns during secure voice-over-IP (VoIP) calls or video conferencing sessions may reveal the start and end times of conversations, as well as potential pauses or interruptions. While this is not a direct content leak, it still poses a privacy concern, particularly in environments where the mere fact that a communication occurred could be sensitive.

8. Conclusion and future work

8.1. Conclusion

In this study, we have introduced an efficient approach to intrusion detection by utilizing the Interpacket Arrival Time (IAT) as a key metric for initial threat filtering. We emphasize that this method is not intended to replace more sophisticated models but rather to serve as a first line of defense that reduces the workload of these advanced systems. Our findings indicate that focusing on the IAT feature allows for the creation of a lightweight, preliminary filtering system that significantly reduces the computational demands on more resource-intensive models. Through extensive testing on datasets such as CICIoT2023, CIC-IDS-2017, and UNSW-NB15, we validated the effectiveness of this approach, achieving high detection accuracy while drastically cutting down on training and prediction times. This method enhances the overall efficiency of intrusion detection systems and offers a practical solution for real-time applications where resource constraints are a critical concern.

8.2. Future work

Building on the promising results of this study, we identify several key areas for future research:

- **Privacy Implications:** While our study highlights the efficiency of using IAT for intrusion detection, future research should explore the privacy implications of this approach. It is essential to ensure that leveraging IAT does not inadvertently expose sensitive network behaviors, particularly in environments where privacy is paramount.
- **Adversarial Robustness:** We acknowledge the importance of evaluating the robustness of our models against adversarial attacks. Although our current datasets naturally include a variety of perturbed IAT values, a focused study on adversarial examples will provide deeper insights into the resilience of our approach under adversarial conditions.
- **Ablation Studies:** While this study concentrated on the effectiveness of the IAT feature within our proposed framework, future work could include detailed ablation studies. These studies would provide additional insights into the contributions of different features and their combinations, helping to refine the filtering system further.
- **Real-World Application and Deployment:** To bridge the gap between research and practical deployment, future work should focus on applying this approach to live network environments. This would involve adapting our models to handle real-time data streams and evaluating their performance in diverse, dynamic network conditions.
- **Extended Model Comparison:** Future research should also consider comparing the proposed ap- proach against a broader range of machine learning models, including more recent and advanced techniques. This comparison would help to further validate the robustness and efficiency of using IAT as a primary feature for intrusion detection.
- **Exploration of Other Time-Based Features:** While this study focused on IAT, other time- based features may also offer valuable insights for intrusion detection. Future work could explore the integration of these features to enhance the detection capabilities of the proposed filtering system.
- **Experimental Simulation of Adversarial Conditions:** Future work should include controlled experiments that simulate adversarial attacks—such as traffic shaping, timed packet injection, and packet bursting—to quantitatively assess the resilience of the IAT-based detection system. We are currently undertaking initial experiments in this direction.
- **Formalization of the Threat Model:** Future work will focus on refining our threat model by de- veloping formal adversarial scenarios and establishing quantitative performance metrics. This involves defining specific attack vectors—such as low-and-slow attacks, traffic bursts, and timing manipula- tions—and creating a simulation environment to mimic real-world adversarial behaviors. By system- atically evaluating the system's performance against these defined threats, we aim to gain deeper insights into the efficacy and limitations of our IAT-based detection approach.
- **Adversarial Validation:** We recognize the need to validate our approach against adversarial attacks. Future studies will focus on designing and conducting experiments that simulate various adversarial scenarios, thereby providing quantitative insights into the system's robustness and informing potential improvements.

By addressing these areas, future research can build upon the foundation laid by this study, further improving the effectiveness and applicability of time-based metrics for intrusion detection in modern network environments.

Compliance with ethical standards

This research strictly adhered to all ethical guidelines and protocols pertinent to cybersecurity and data analysis studies.

Competing interests

The authors declare that they have no competing interests. All authors confirm that there are no financial affiliations or any other personal conflicts of interest that could have influenced the work reported in this study.

Research Data Policy and Data Availability Statements

Data Accessibility

The datasets used in this study are publicly available at:

https://www.kaggle.com/datasets/subhajournal/iotintrusion/data, https://www.kaggle.com/datasets/dhoogla/unswnb15, https://www.kaggle.com/datasets/chethuhn/network-intrusion-dataset

Data Restrictions

There are no restrictions on the datasets utilized in this study, ensuring transparency and reproducibility.

CRediT authorship contribution statement

Onur Sahin: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Suleyman Uludag:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data is publicly available.

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